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Cell Biology

Topic No 1. : Introduction

Course Outline

Atom→ Molecules→ Cell Structures→ Cells

Cell Functions:

Propagation/Heredity

Expression Regulation

Cell Communication

Applied Cell Biology

Course Objective:

Understand the molecular logic of the "living state"

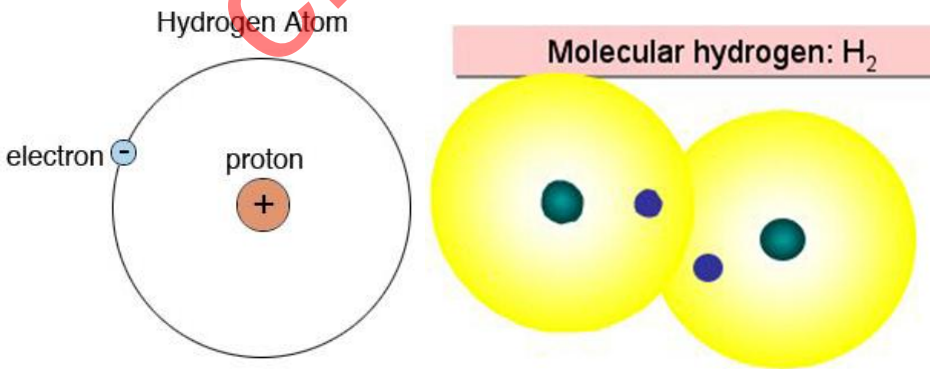
How a molecule's function arises from its structure,

How this information is conveyed to the rest of the cellular machinery

System Manipulation

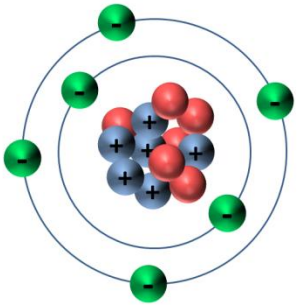
Topic 2: Composition of Matter

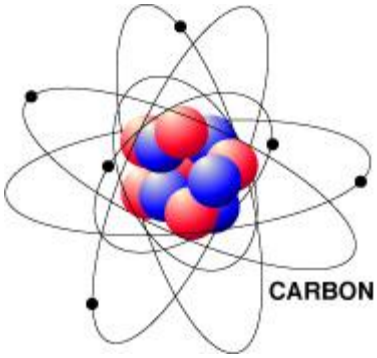
1. **Atomic Structure**



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Carbon Atom





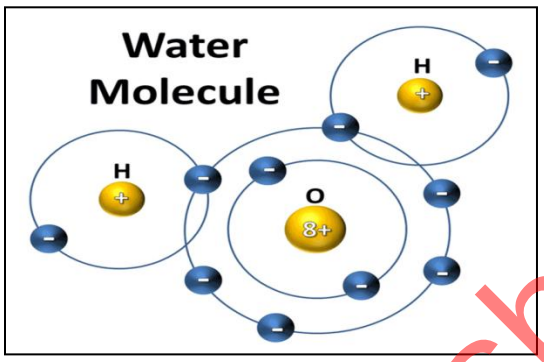


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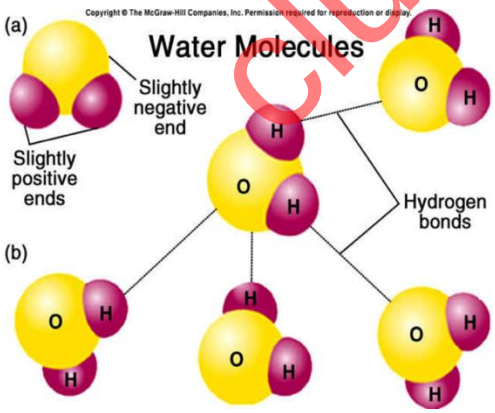
Electronegativity:

The tendency of an atom to attract electrons towards itself.



Polarity of H2O:

Results in Hydrogen Bonding

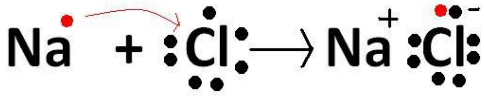


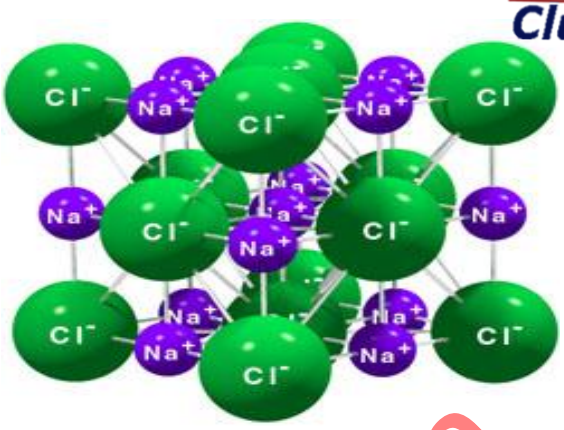
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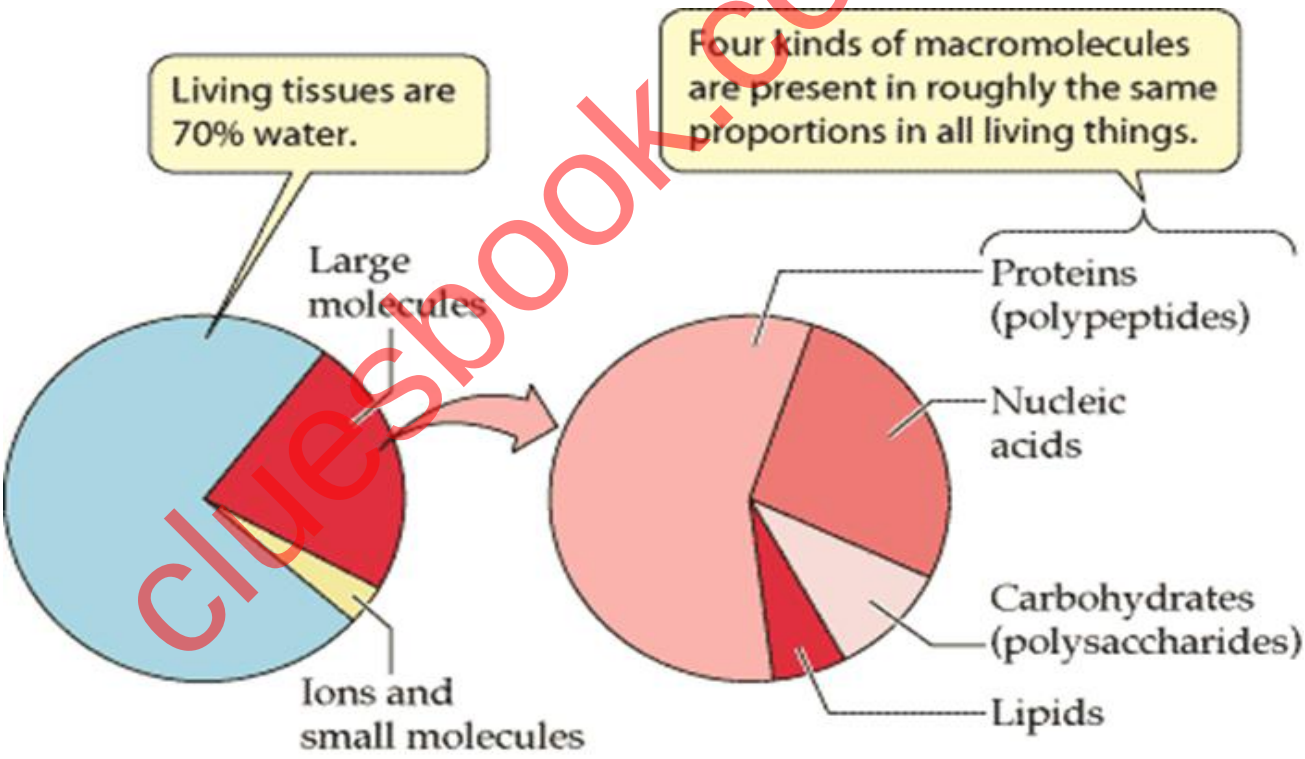
Ionic Bond:





Topic No 3 : Molecules of Life

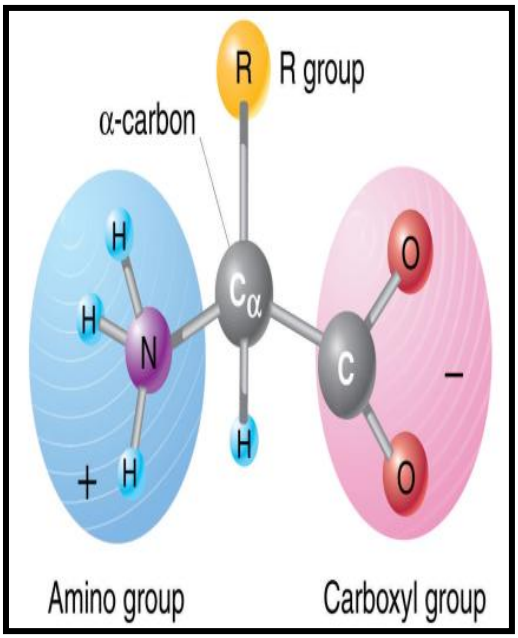
1. **Composition of Life**

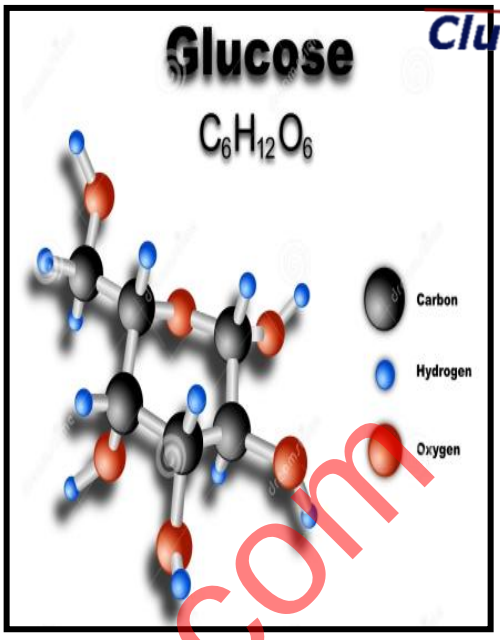


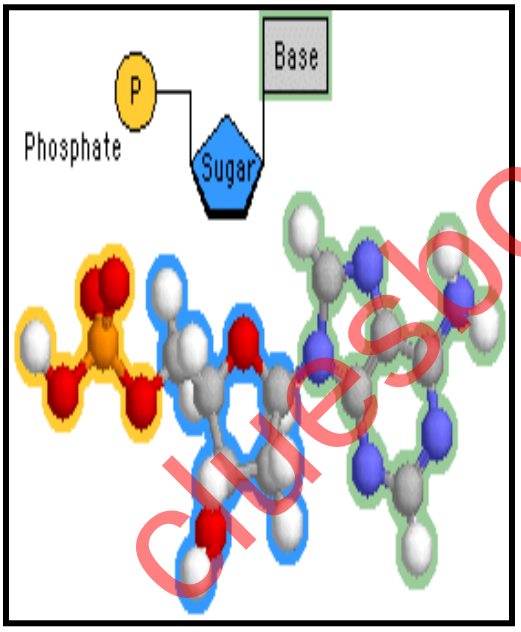
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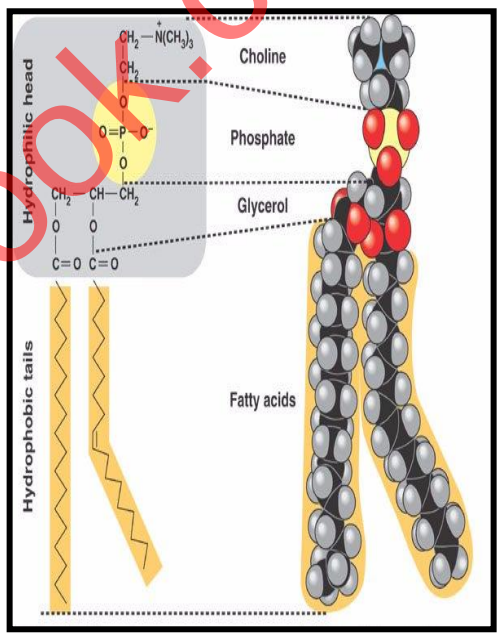
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2. Building Blocks of Life









3. Polymers & Monomers

Macromolecules are polymers of smaller monomers molecules

Condensation reactions: Monomers join H_2O Removal

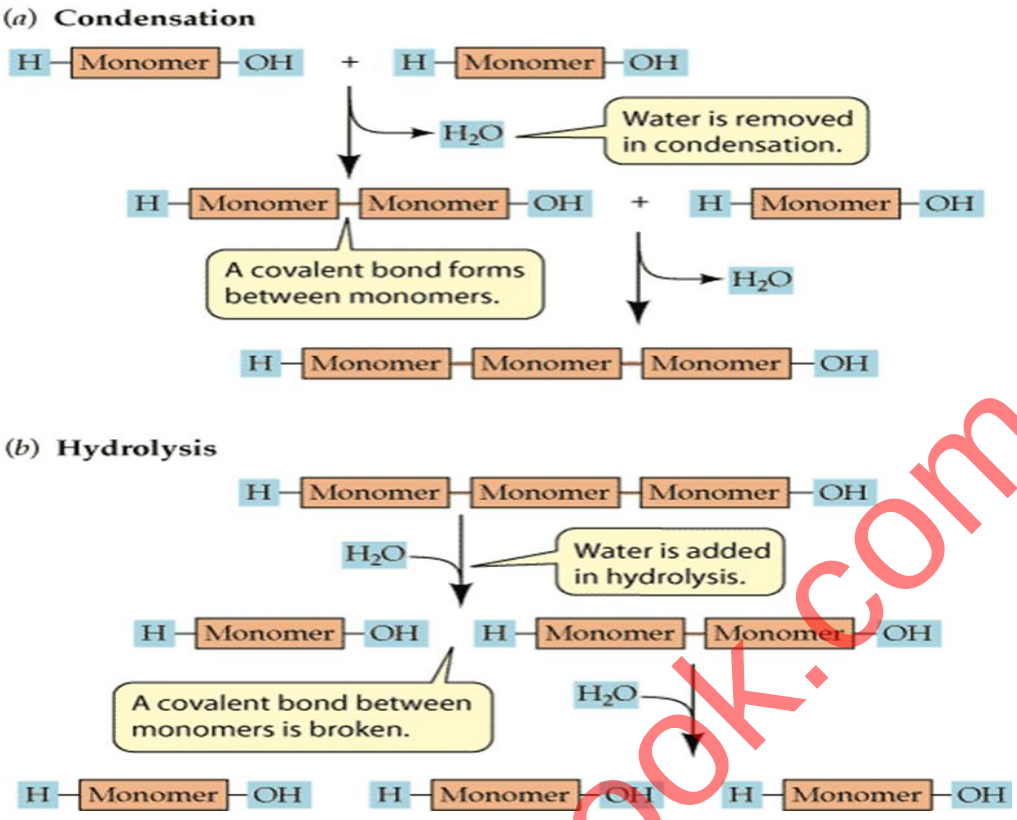
Hydrolysis reactions: Use water to break polymers into monomers.

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4. Condensation & Hydrolysis



Topic No 4. Proteins: Polymers of Amino Acids

1. Protein Functions

Most Abundant Polymer in the cell

Support, protection, catalysis, transport, defense, regulation & movement

Do not store genetic information

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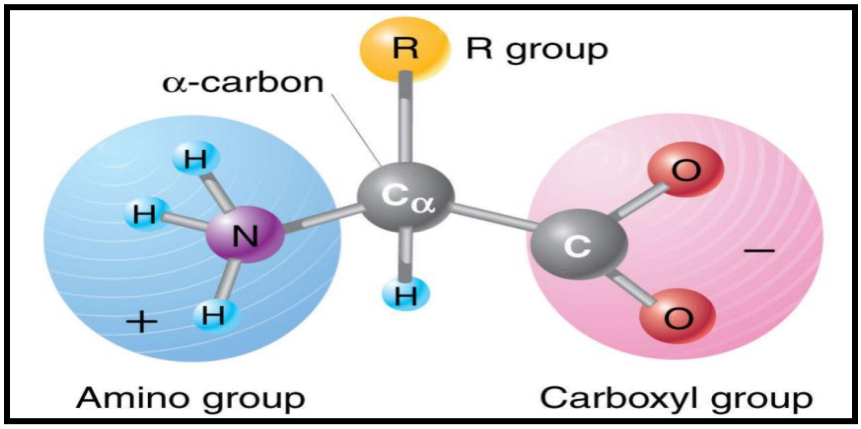
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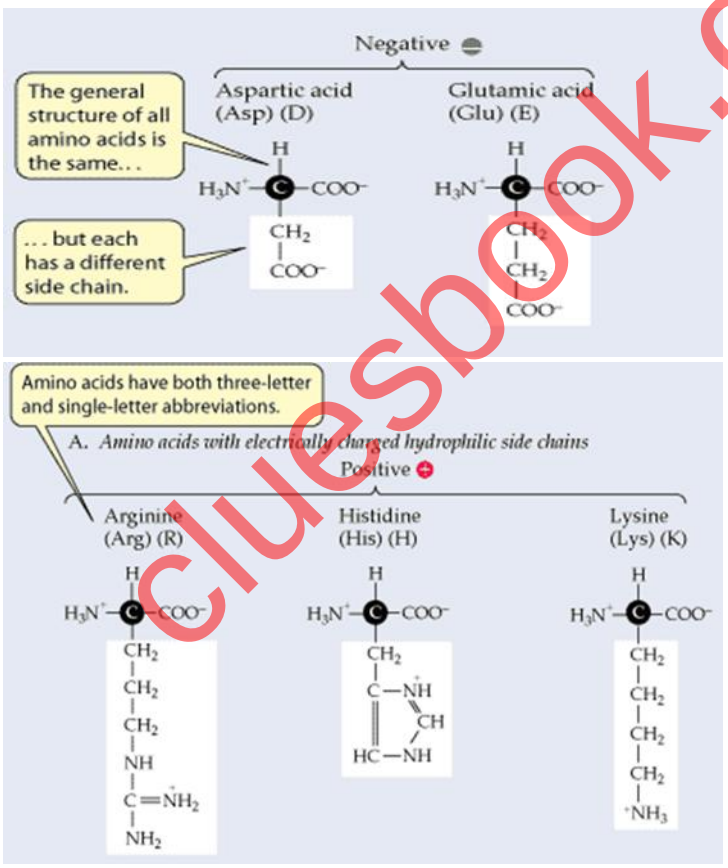
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2. **Amino Acid Structure**

The side chains or R groups give different properties to each of the amino acids



3. **Charged R Groups**



Charged, polar, or hydrophobic

Special case cysteine SH groups can form disulfide bridges

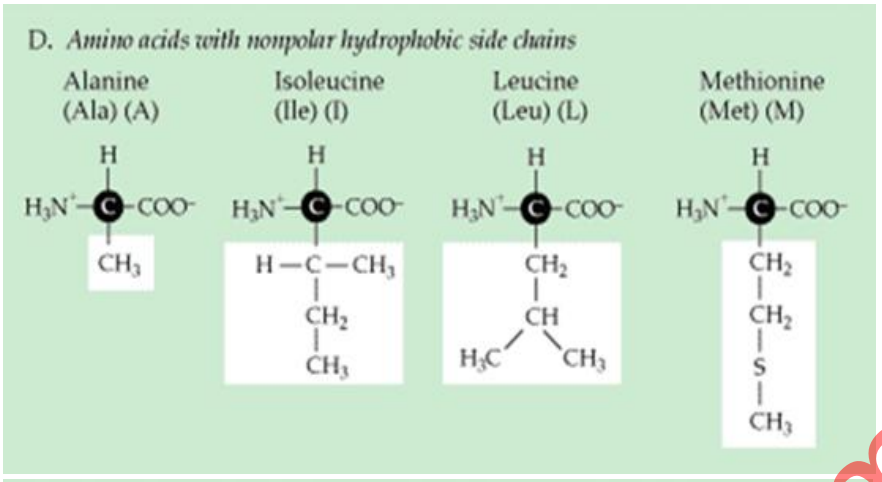
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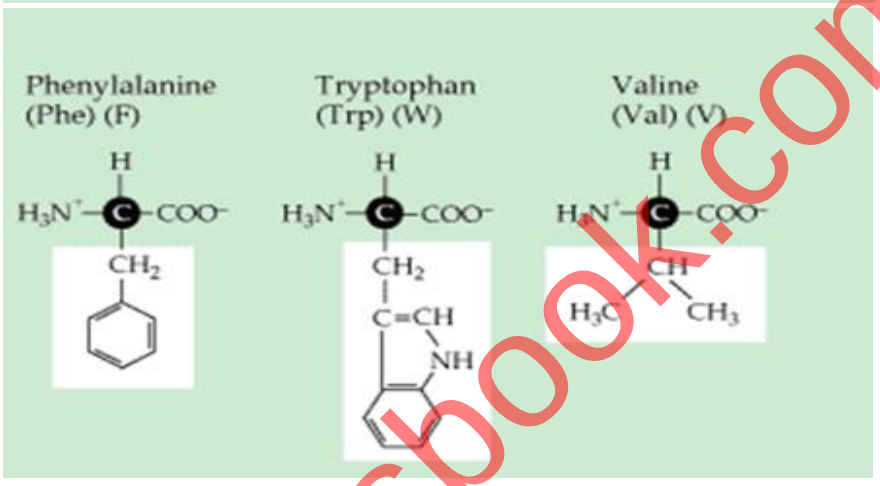
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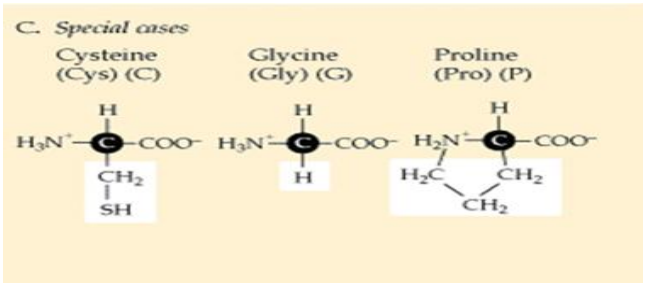
4. **Nonpolar R Groups**





5. **Polar & Special R Groups**





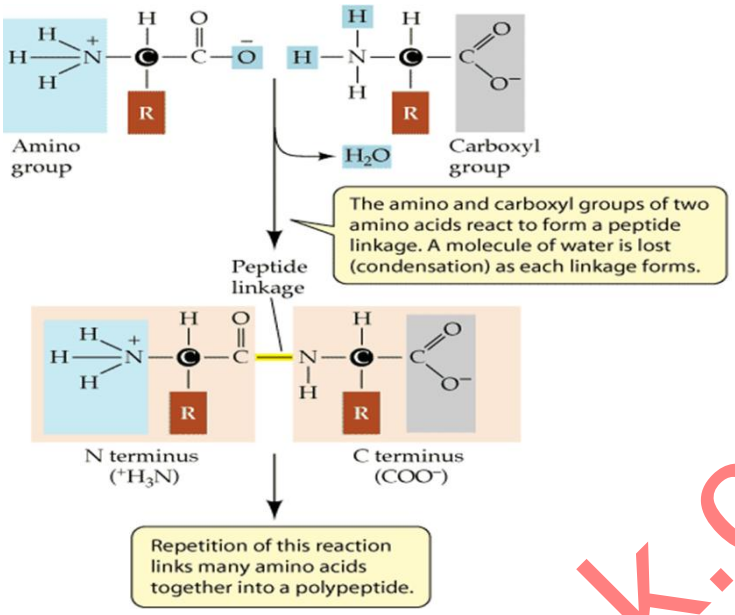
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Topic No. 5 :Proteins Structure

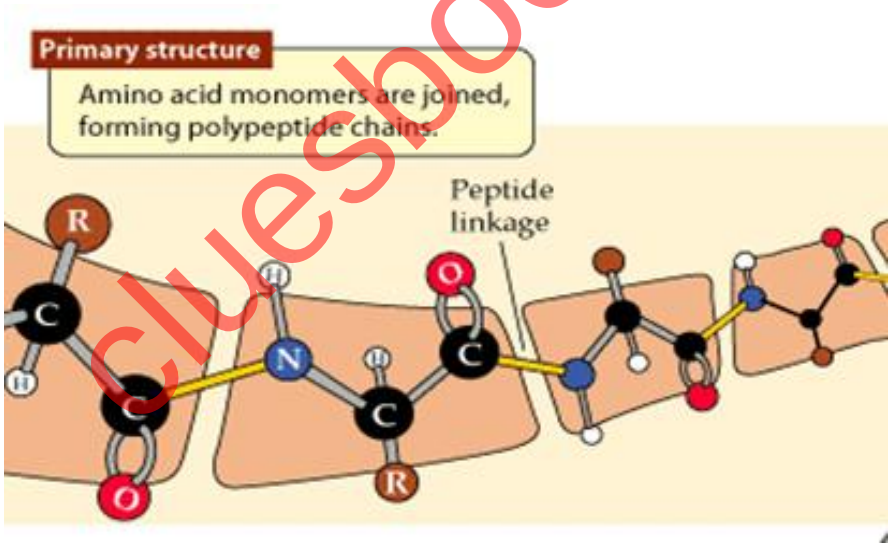
1. **Proteins: Polymers of Amino Acids**

   The Peptide Bond



2. **Proteins: Structure**

   Primary Structure



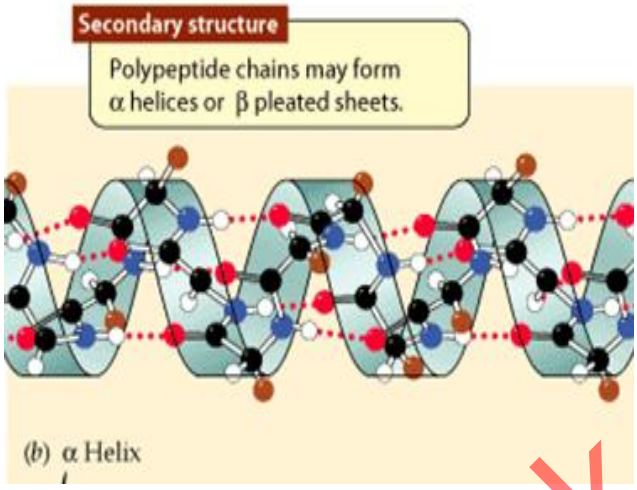
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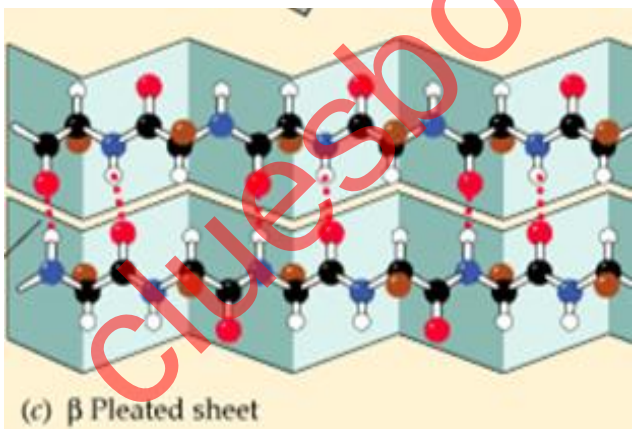
Sequence of amino acids bonded by peptide linkages (Diversity 20n)

Secondary Structure

i. **α helices (hydrogen bonds between amino acid residues)**



ii. **β pleated sheets (hydrogen bonds between amino acid residues)**

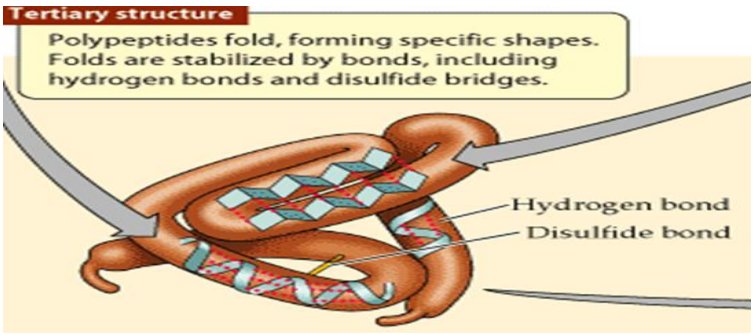


Tertiary Structure

Generated by bending and folding of the polypeptide chain

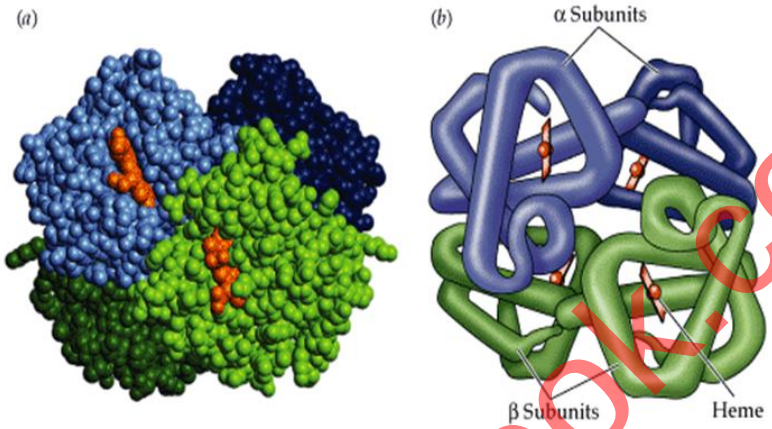
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Quaternary structure

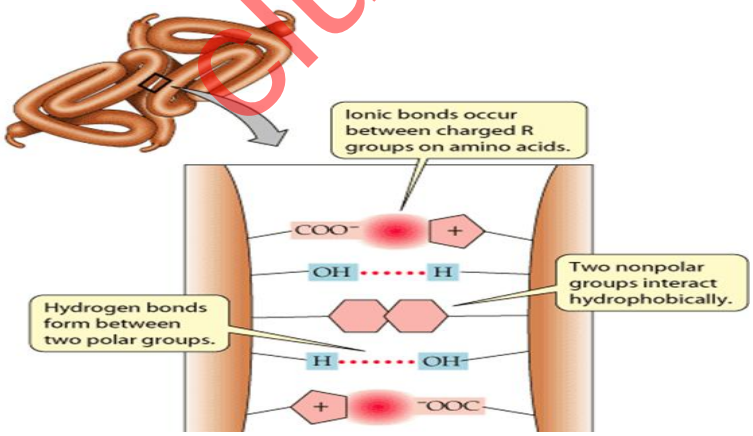
Arrangement of multiple folded protein molecules in a multi-subunit complex.



Topic 6 : Protein Interactions

1. Protein Interactions

Weak interactions and 3D structure

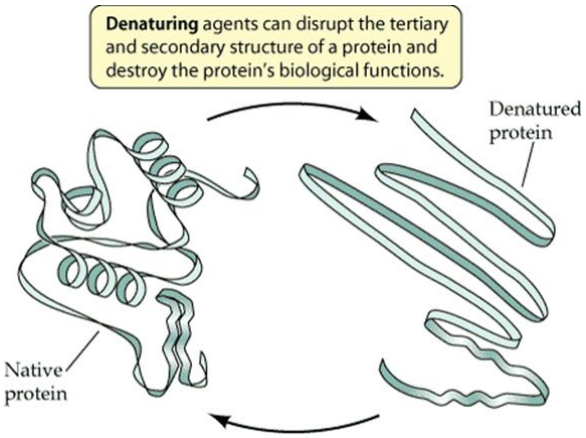


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2. **Protein: Denaturation**

Heat, pH change: Tertiary and secondary structure as well as biological function



Topic 7. Sugar Polymers

Carbohydrates are sugar polymers

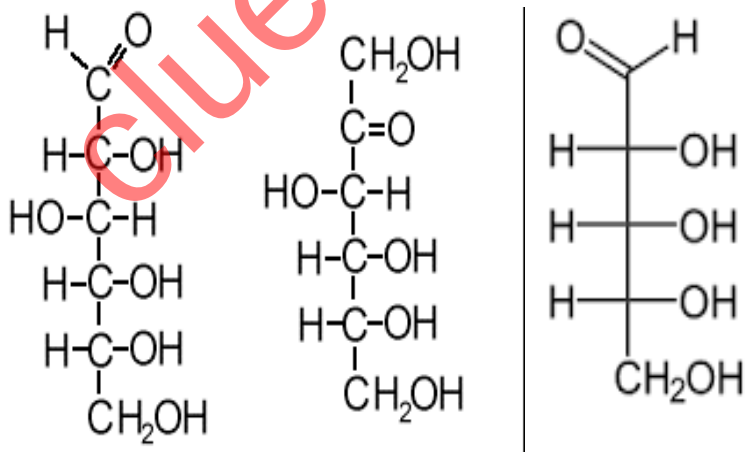
They act as source of energy that can be transported

They also have structural role.

i. Monosaccharides
ii. Disaccharides
iii. Oligosaccharides (3-20)
iv. Polysaccharides

1. **Monosaccharides:**

Produced by plants, all living cells have glucose.



Glucose Ribose Fructose

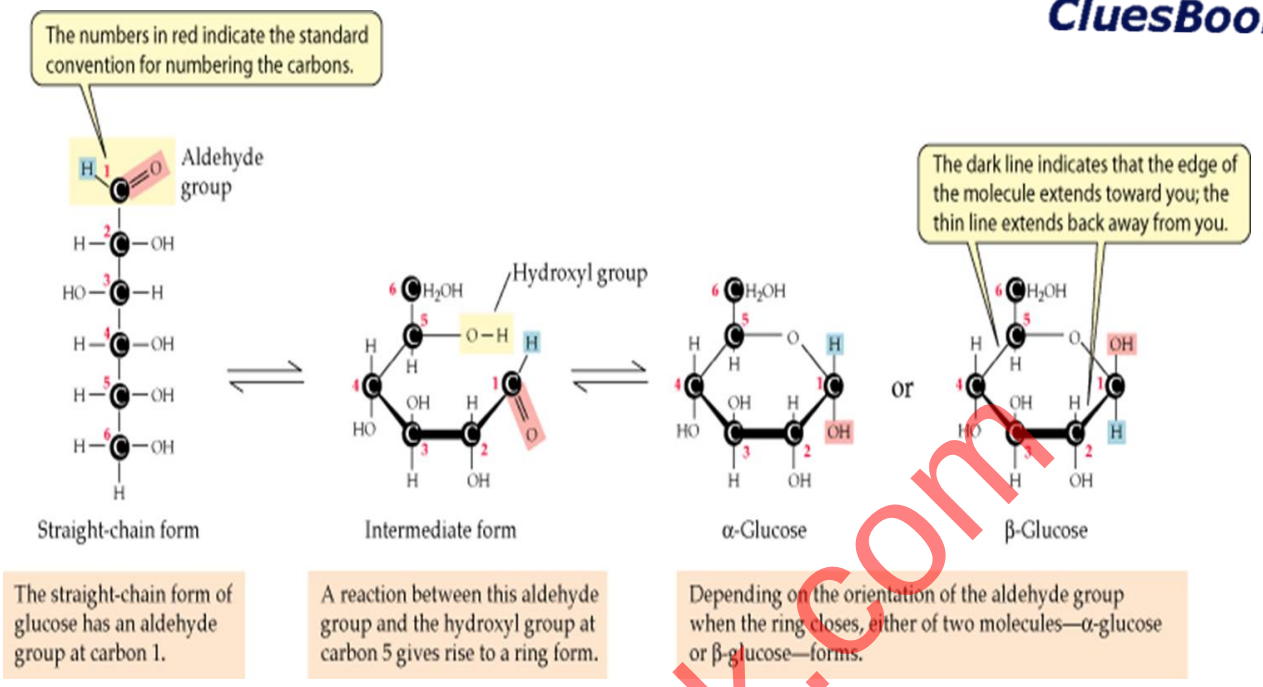
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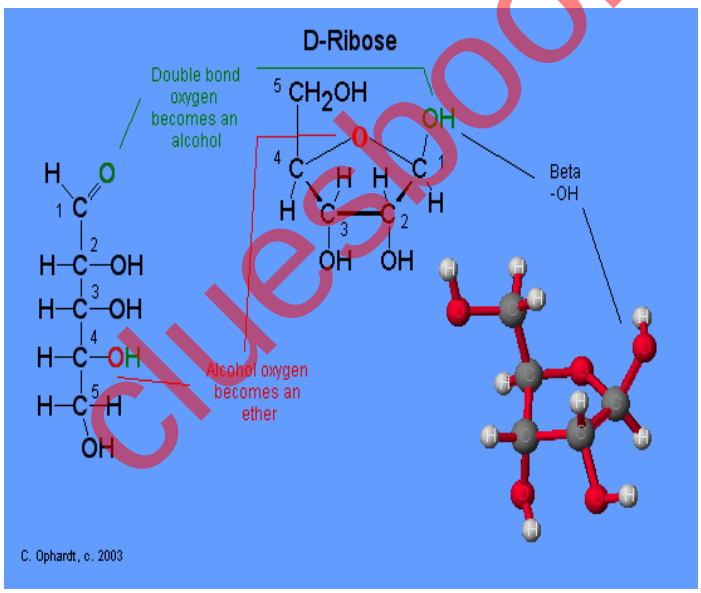
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2. Glucose Two Forms



3. Monosacchrides Ribose



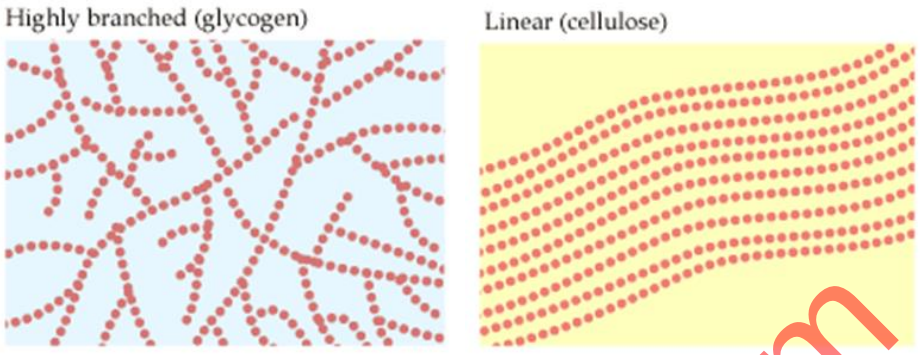
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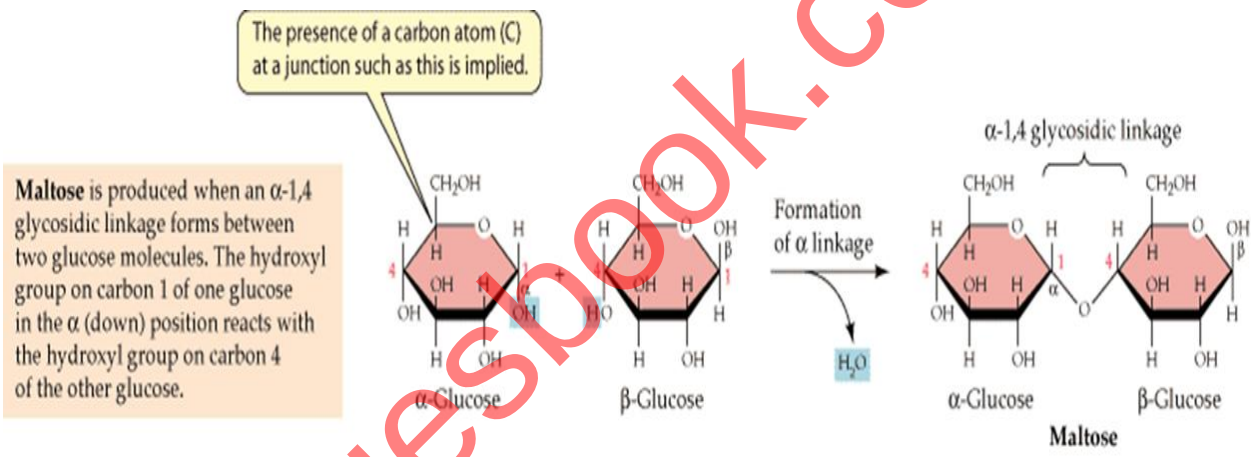
Topic No 8 : Glycosidic linkages

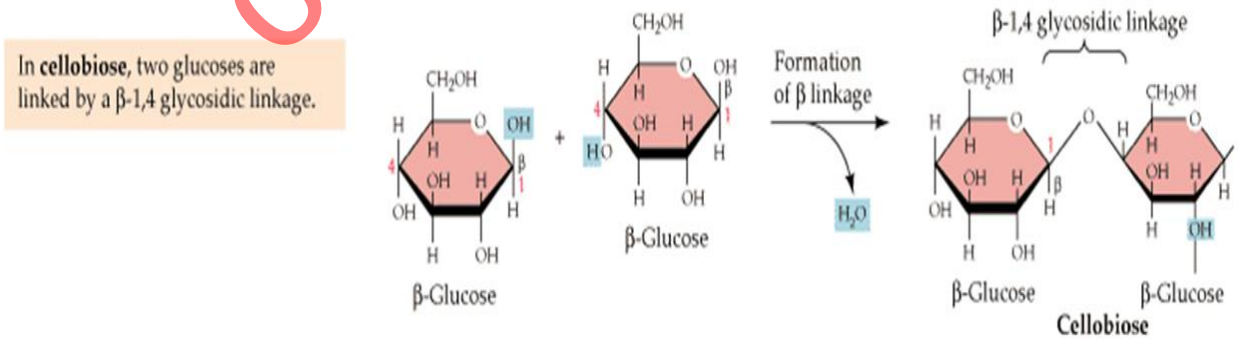
1. **Carbohydrates**

Disaccharide α or β orientation



2. **Disaccharide α or β orientation**





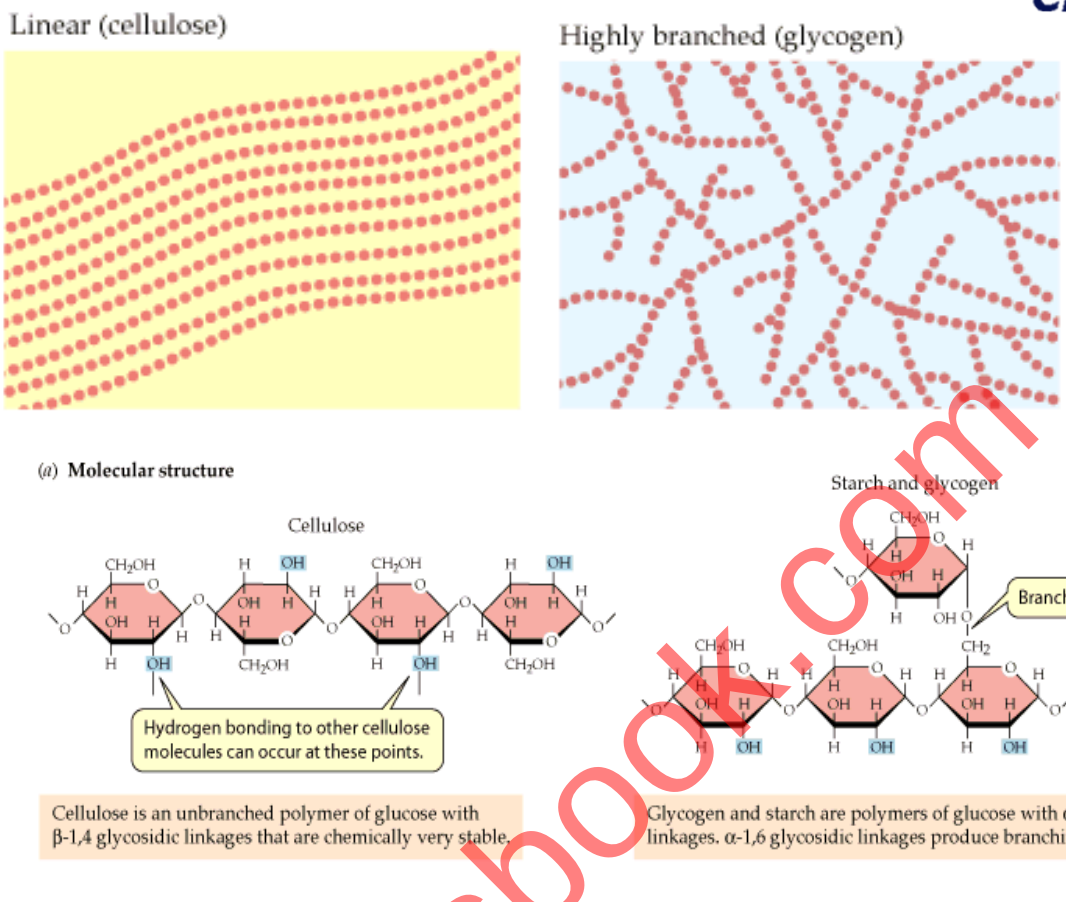
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3. Cellulose & Glycogen



Cellulose is an unbranched polymer of glucose with β-1,4 glycosidic linkages that are chemically very stable.

Glycogen and starch are polymers of glucose with α-1,4 glycosidic linkages. α-1,6 glycosidic linkages produce branching at carbon 6.

4. Cellulose & Starch



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Topic 9 Lipids

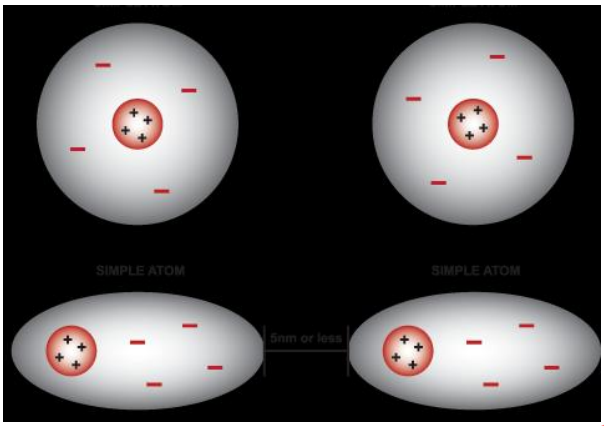
Lipids are Nonpolar Hydrocarbons

Week Bonds

Hydrophobic molecules

Water insoluble

Aggregate together (by hydrophobic and Van der Waals force)



Functions

Store energy

Phospholipids membranes

Carotenoids & Light

Hormones and vitamins

Animal fat is insulator

Insulation of nerves

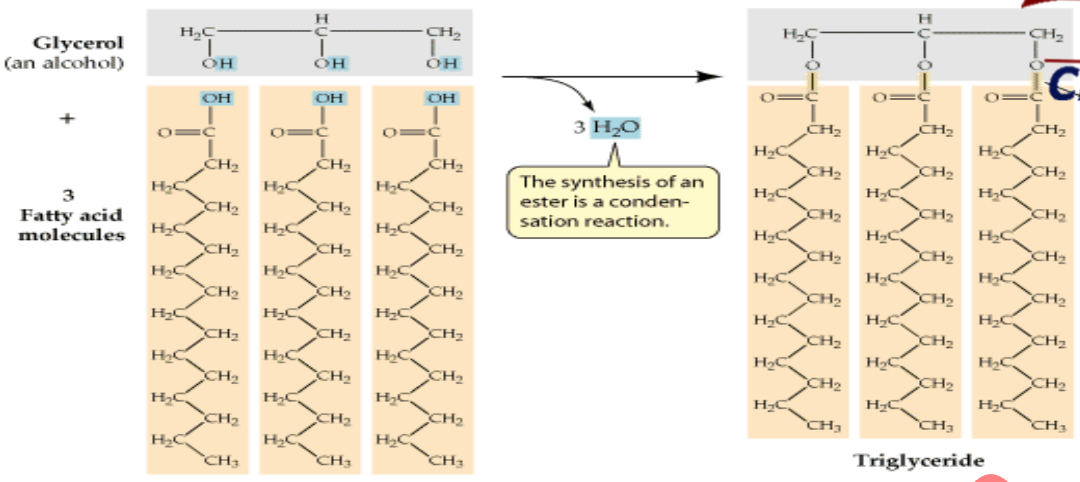
Water repellant for skin, fur and feathers

Lipids: Synthesis

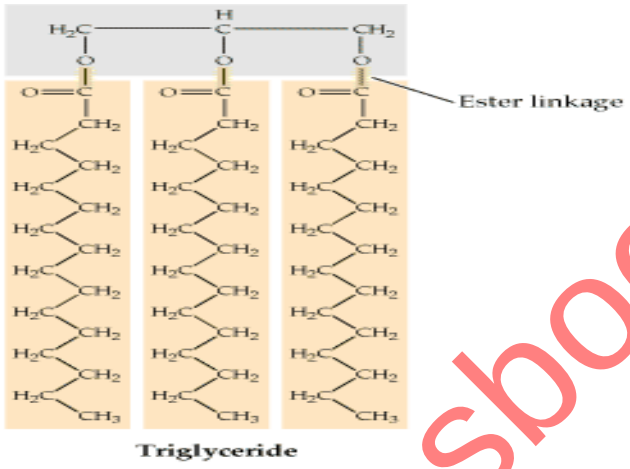
Fats & oils: Triglycerides: (covalent ester linkage vs Hydrophobic bond)

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Lipids: Synthesis



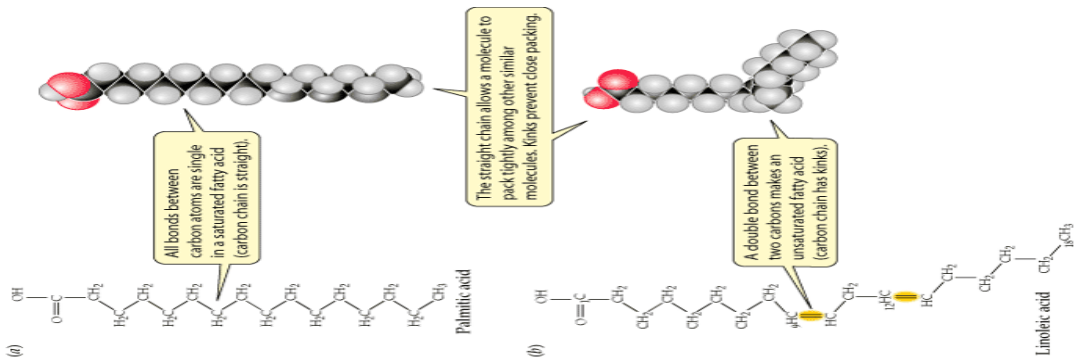
Topic 10 Phospholipids

1. **Lipids: Saturated & unsaturated**

   Fats (L) & Oils (S)

   Saturated fatty acids have a hydrocarbon chain with no double bonds.

   The hydrocarbon chains of unsaturated fatty acids have one or more double bonds that bend the chain, making close packing less possible.

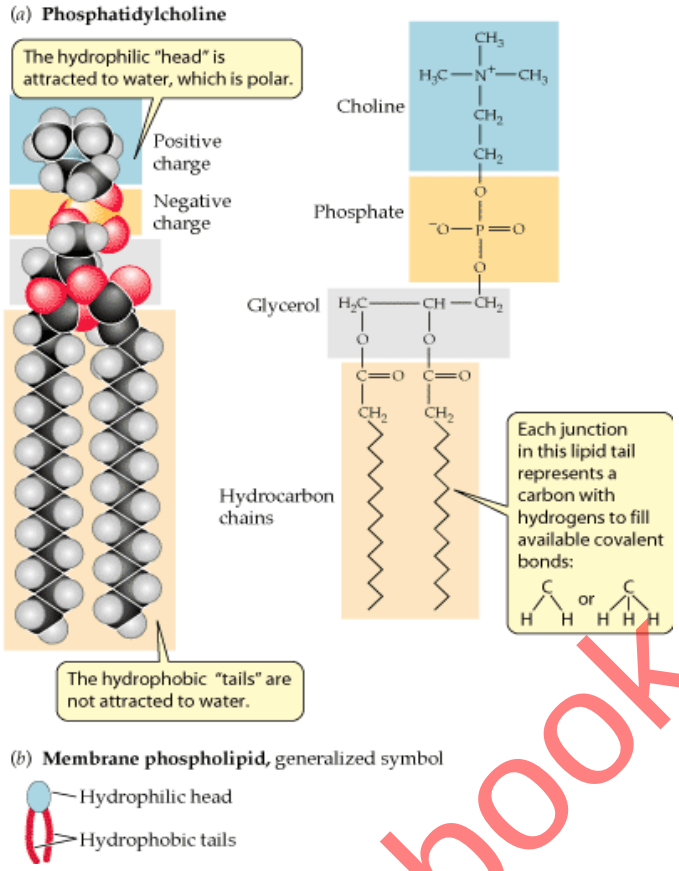


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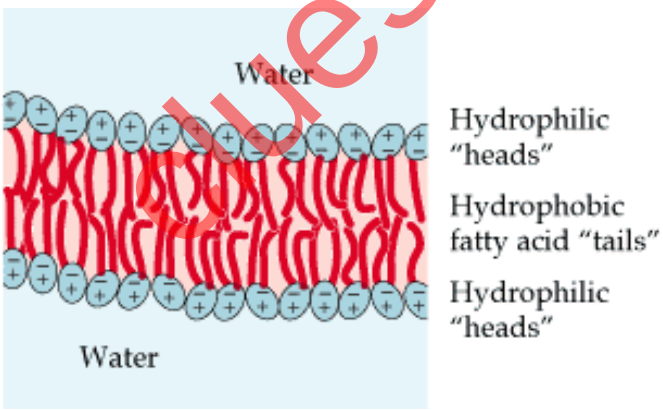
2. **Phospholipids**

Amphipathic Molecule



3. **P-lipids: Aqueous Environment**

The Bilayer : Two molecules thick head groups directed outward.

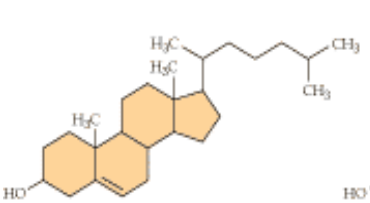


4. **Lipids: Cholesterol**

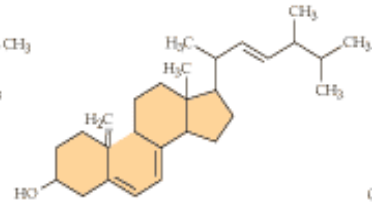
Vitamins & Signals

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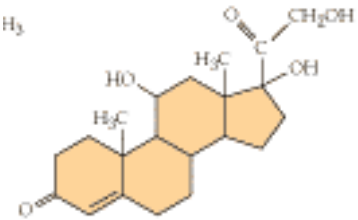
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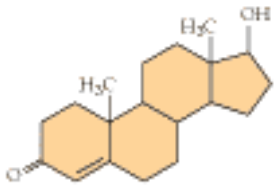
Cholesterol is a constituent of membranes and is the source of steroid hormones.



Vitamin D_2 can be produced in the skin by the action of light on a cholesterol derivative.



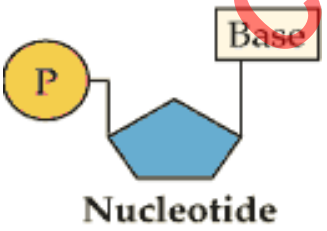
Cortisol is a hormone secreted by the adrenal glands.

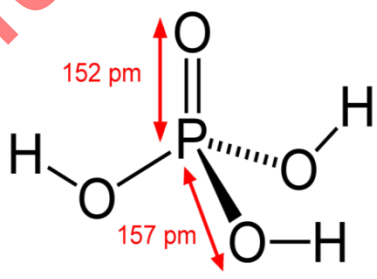


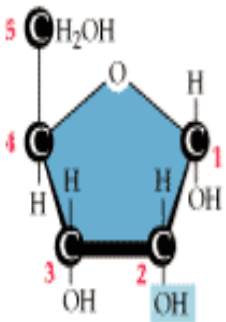
Testosterone is a male sex hormone.

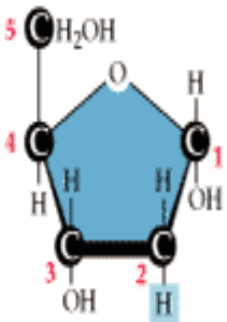
Topic 11 Nucleic Acids

Types are DNA & RNA





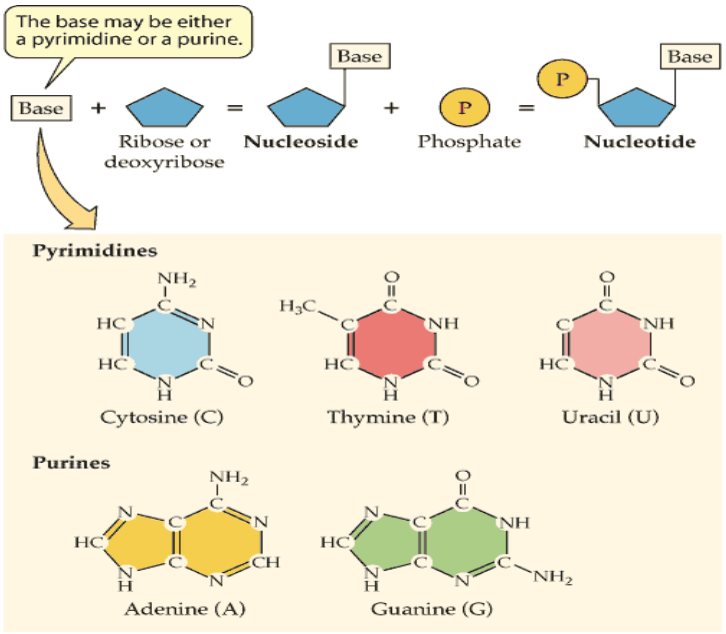




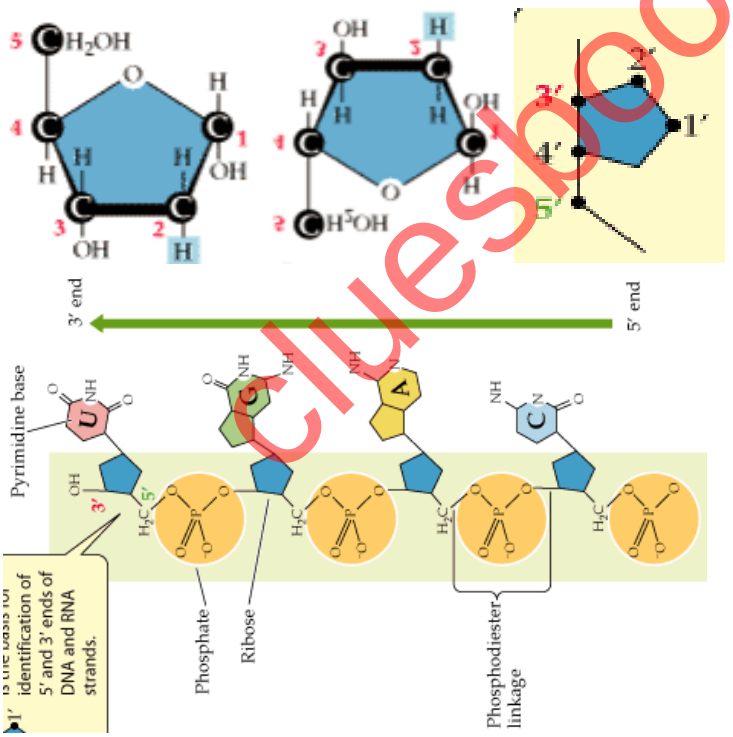
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The Basis



Prime Ends (3'& 5')



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Topic 12 RNA AND DNA

RNA VS DNA

3.3 Distinguishing RNA from DNA

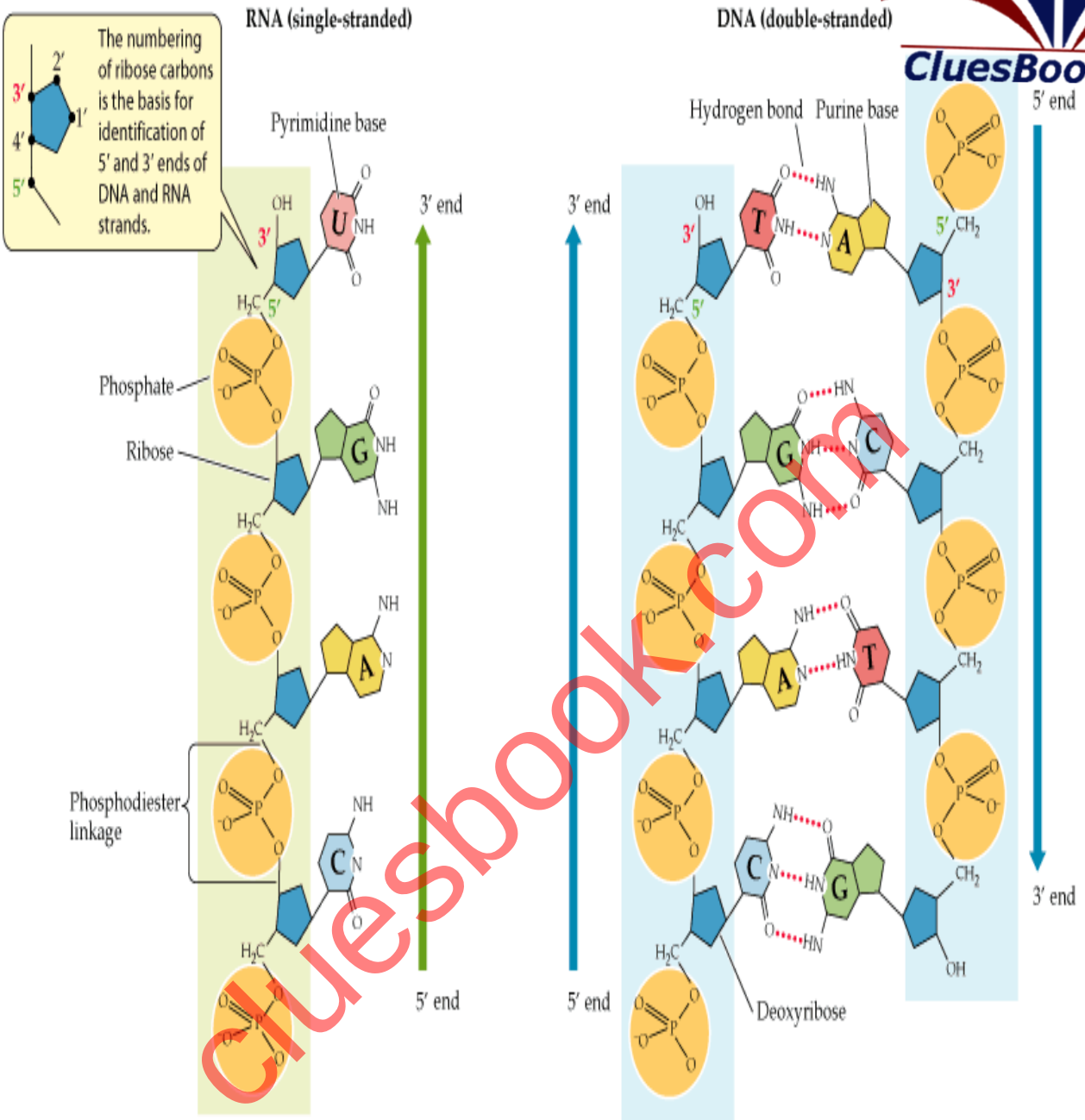
NUCLEIC ACID	SUGAR	BASES
RNA	Ribose	Adenine Cytosine Guanine Uracil
DNA	Deoxyribose	Adenine Cytosine Guanine Thymine

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In RNA, the bases are attached to ribose. The bases in RNA are the purines adenine (A) and guanine (G) and the pyrimidines cytosine (C) and uracil (U).

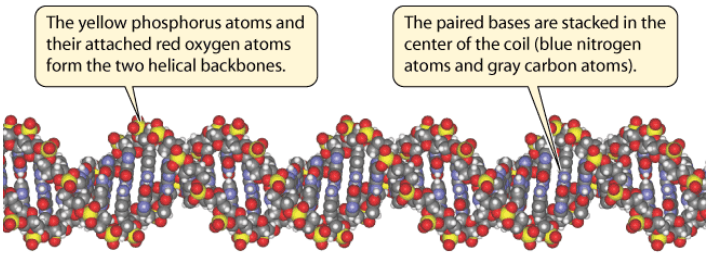
In DNA, the bases are attached to deoxyribose, and the base thymine (T) is found instead of uracil. Hydrogen bonds between purines and pyrimidines hold the two strands of DNA together.

DNA Uniform Width

Information in Sequence not Shape

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Chicken vs. Egg: What Came 1st

Protein or DNA

Genetic information codes for enzymes

Enzymes are required to decipher genetic information and replicate it

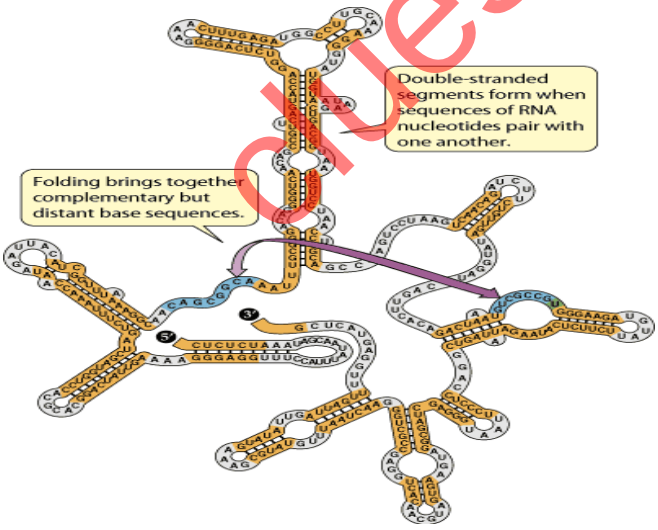
RNA Genetic Material & Enzyme

Many viruses use RNA as their hereditary material

RNAs can achieve chemical catalysis, like enzymes: ribosome the active site is composed entirely of RNA (peptide bond formation)

RNA: Information in Sequence and Shape

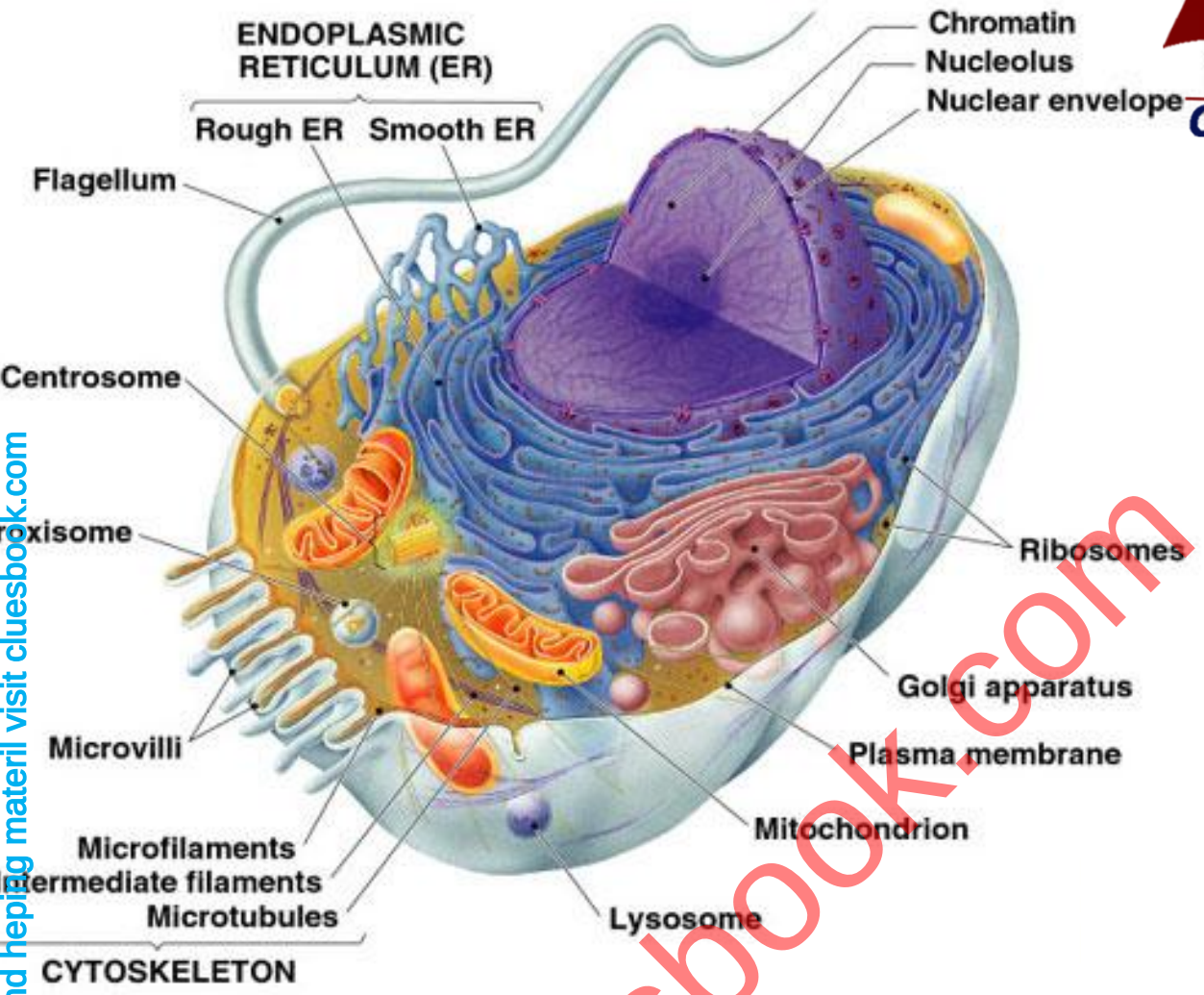
Why Then DNA?



Topic 13 Unit of Life

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Not a random collection of molecules

Life is a collection of macromolecules that can perform unique functions because they are enclosed in structural compartment that provides consistency (homeostasis).

Topic 14 Journey into the Cell

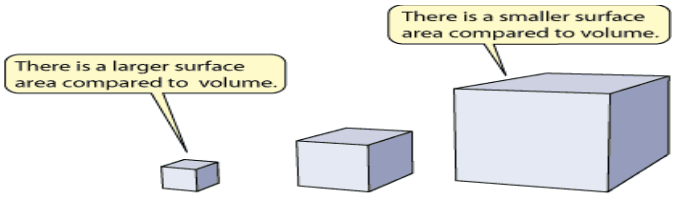
Topic 15 Size Matters

Cell Structure

Cells are small to maintain large surface area to volume ratio

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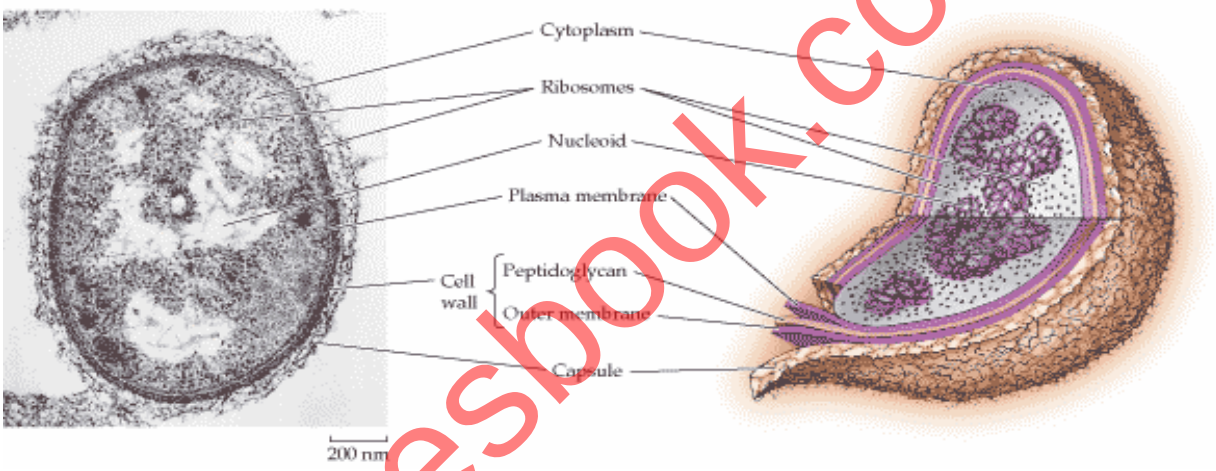
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	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

Prokaryotic Cells

No membrane enclosed internal compartments. Plasma membrane regulates traffic (barrier).Nucleoid region contains DNA.



Special Features

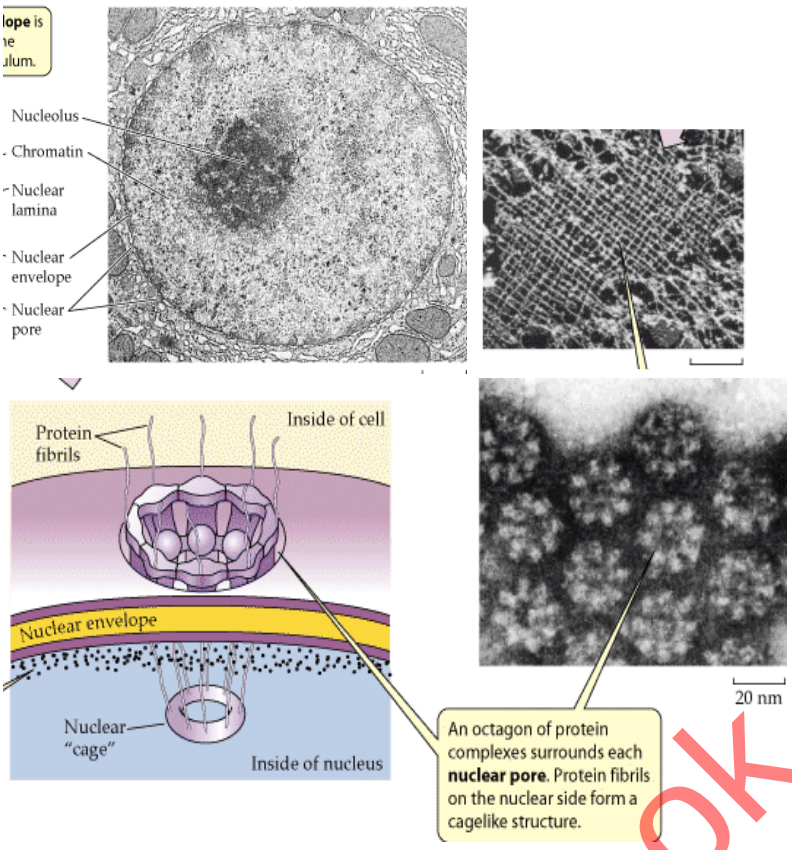
Cyanobacteria Chlorophyll containing folds of plasma membrane, other have mesosomes (energy).Some have actin like filaments and other have Flagella made-up of Flagellin.

Topic 16 The Nucleus

Nuclear EMs

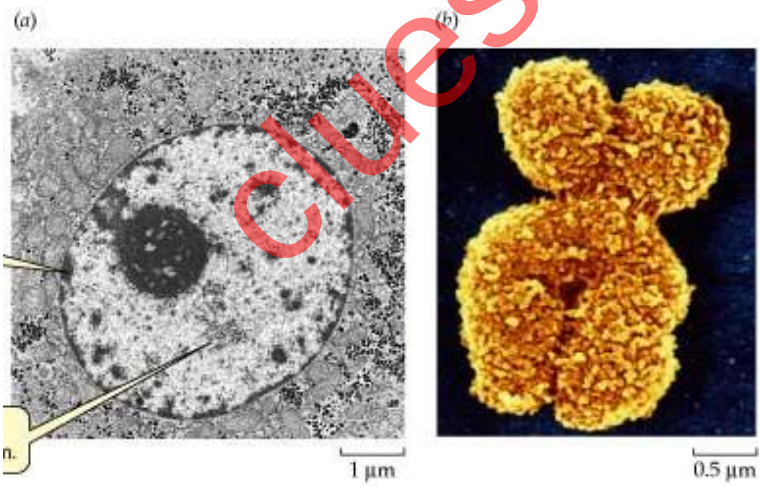
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Chromatin

Euchromatin --- Hetrochromatin --- Chromosome



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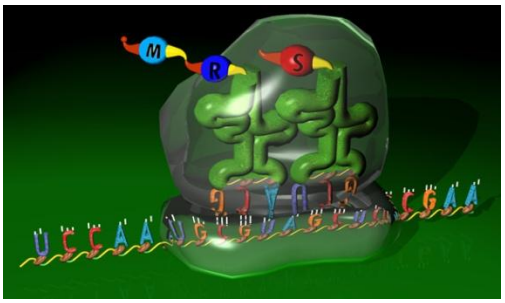
Topic 17 Ribosomes

Ribosome

Protein & RNA

Free or attached to ER involved in protein synthesis.

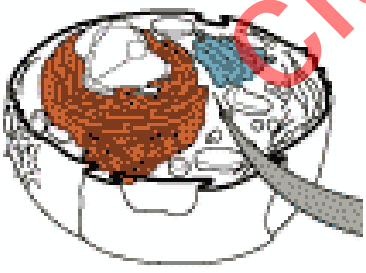
Mitochondria & Chloroplasts.

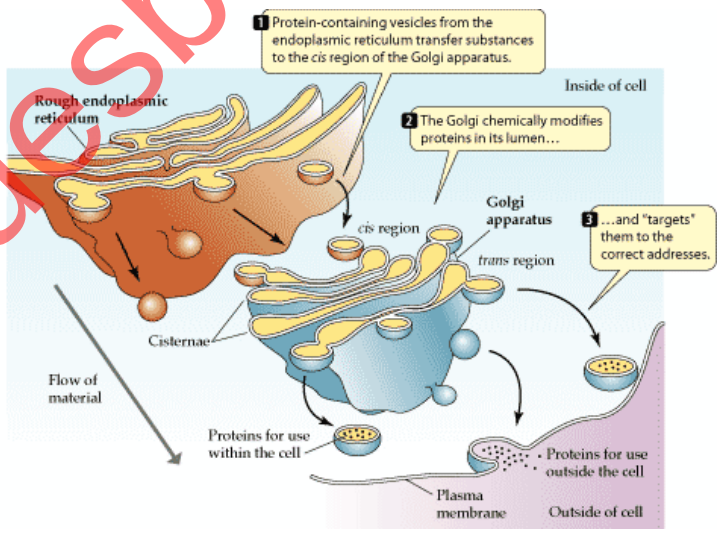


Topic 18 The Endoplasmic Reticulum

The Endoplasmic Reticulum

RER + SER





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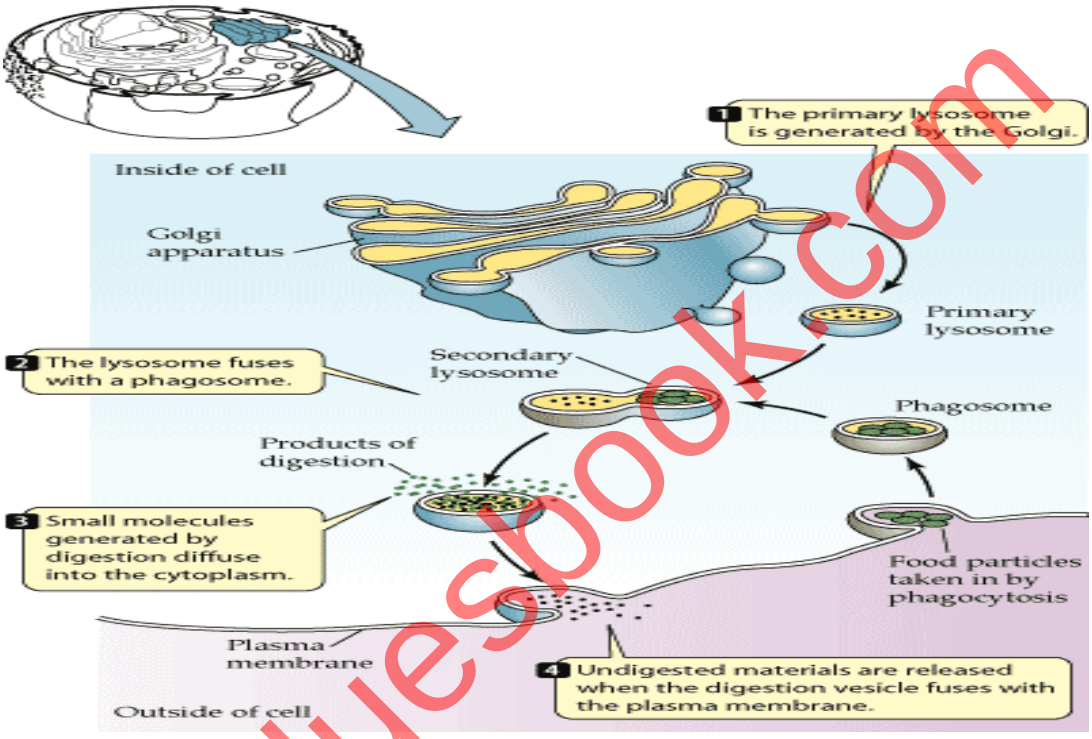
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Topic 19 Golgi and Lysosomes

Golgi

Golgi Apparatus

Receives materials from the rough ER and modifies them. Concentrates sorts and packages proteins and sends them to other destinations.Manufactures poly-saccharides for the plant wall.



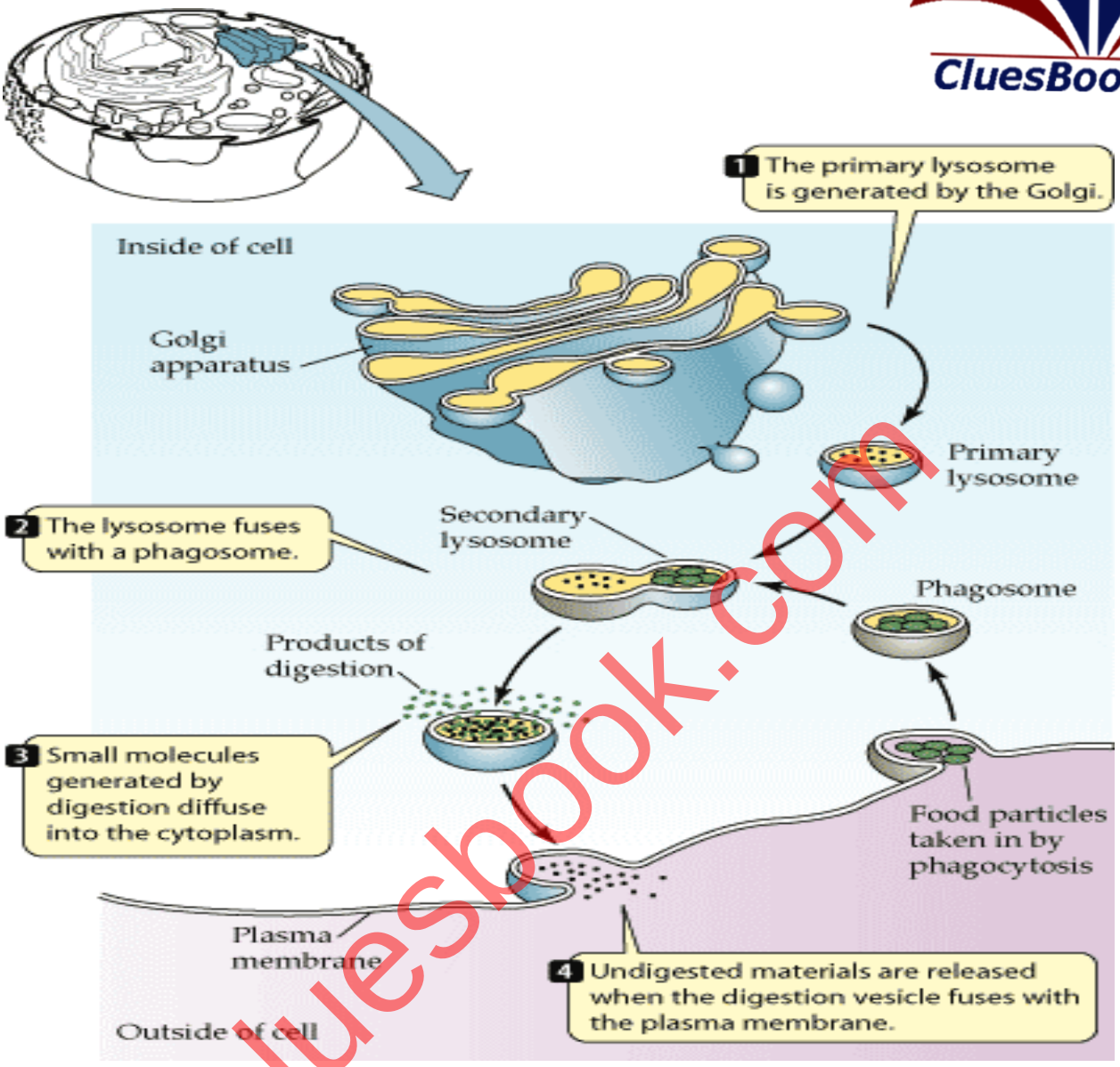
Lysosomes

Garbage Disposal

Contain many digestive enzymes. Fuse with phagosomes (autophagy).Secondary lysosomes, where engulfed materials are digested phagocytosis.

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Topic 20 Mitochondria

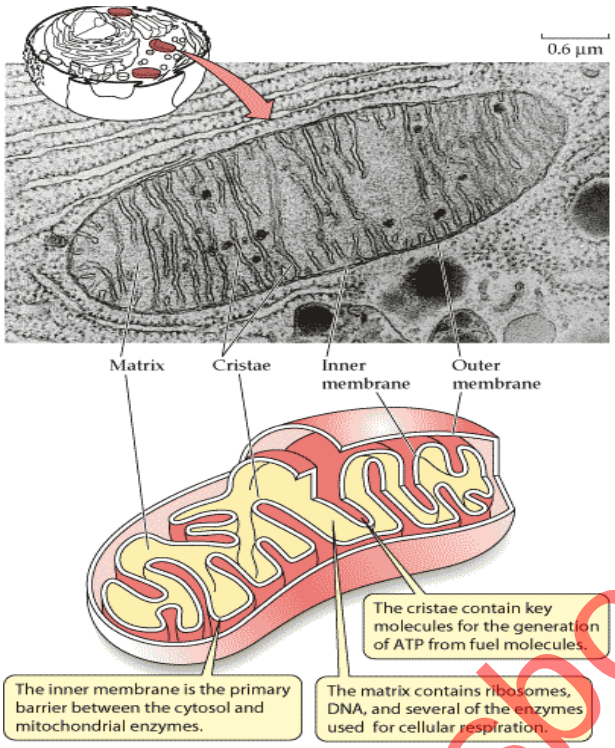
Mitochondria

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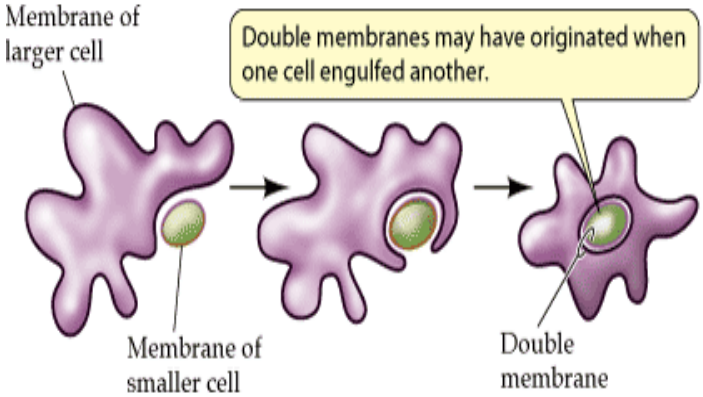
Make ATP Breakdown fuel

Some protists have one liver more than a thousand and egg a few hundred thousand MT. Mitochondria and chloroplasts contain their own DNA and ribosomes for making some of their own proteins



Endosymbiosis

Mutual benefits permitted this symbiotic relationship. Photosynthesis Detoxification of O_2.



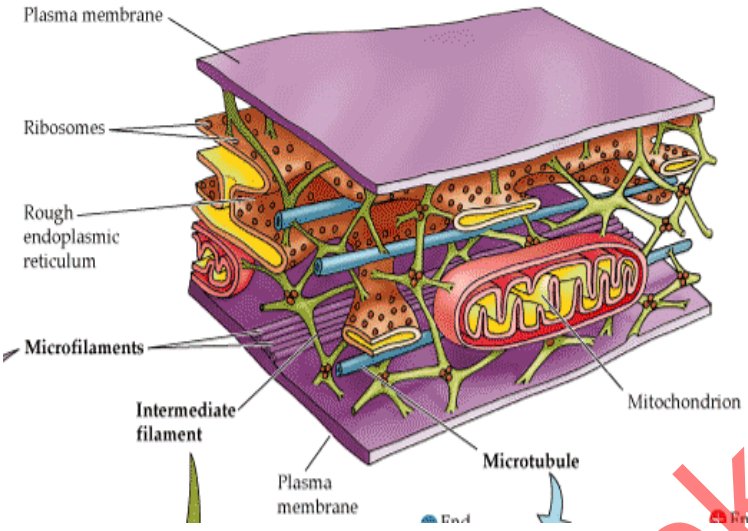
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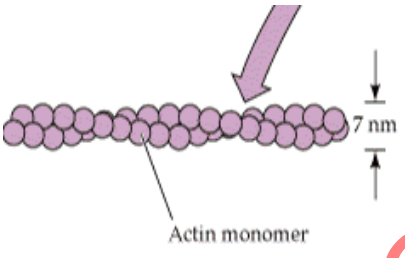
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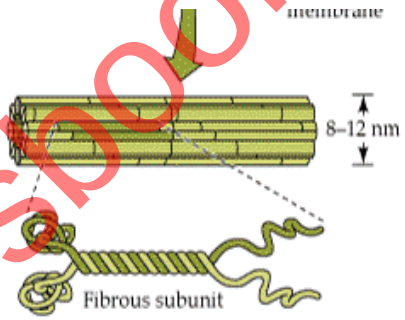
Topic 21 Cytoskeleton-I

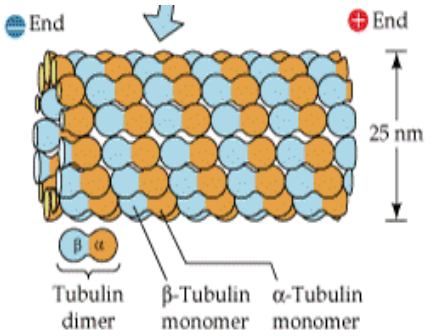
Cytoskeleton

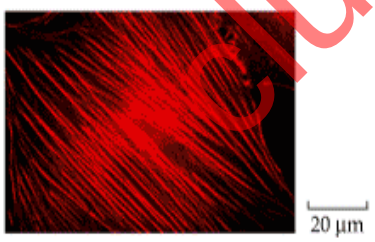
Shape & Function

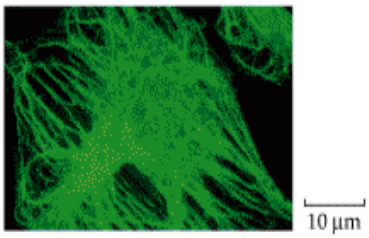


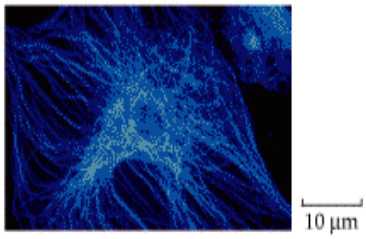












Microfilaments are made up of strands of the protein actin and often interact with strands of other proteins. They change cell shape and drive cellular motion, including contraction, cytoplasmic streaming, and the "pinched" shape changes that occur during cell division. Microfilaments and myosin strands together drive muscle action.

Intermediate filaments are made up of fibrous proteins organized into tough, ropelike assemblages that stabilize a cell's structure and help maintain its shape. Some intermediate filaments help to hold neighboring cells together. Others make up the nuclear lamina.

Microtubules are long, hollow cylinders made up of many molecules of the protein tubulin. Tubulin consists of two subunits, α-tubulin and β-tubulin. Microtubules lengthen or shorten by adding or subtracting tubulin dimers. Microtubule shortening moves chromosomes. Interactions between microtubules drive the movement of cells. Microtubules serve as "tracks" for the movement of vesicles.

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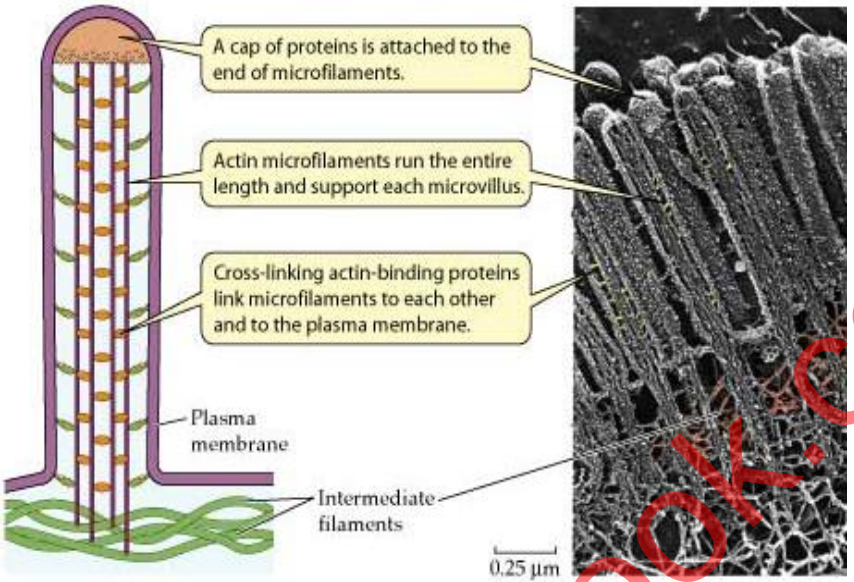
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Actin

Actin (cortical) Stabilizes cell shape. Generates movement local. Muscle contraction. Constriction ring in cell division (Cytokinesis). Cytoplasmic streaming.

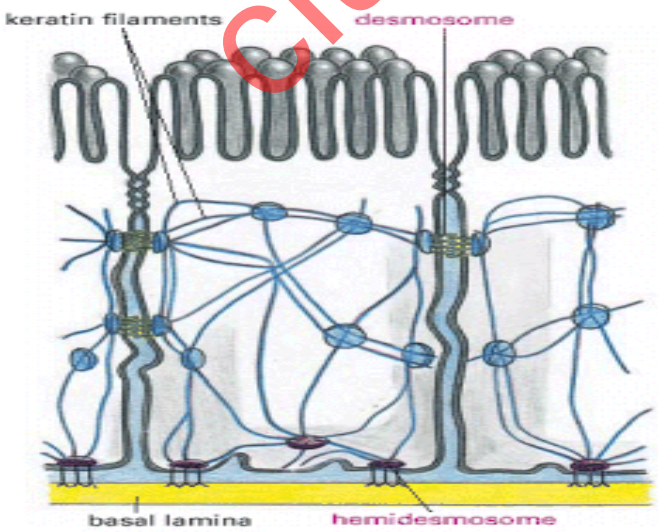
Actin for Support



Topic 22 Cytoskeleton-II

IF

IF stabilize cell structure (hold organelles). Stabilize and maintain tissue rigidity (desmosomes). Laminas in nucleus. Resist tension.



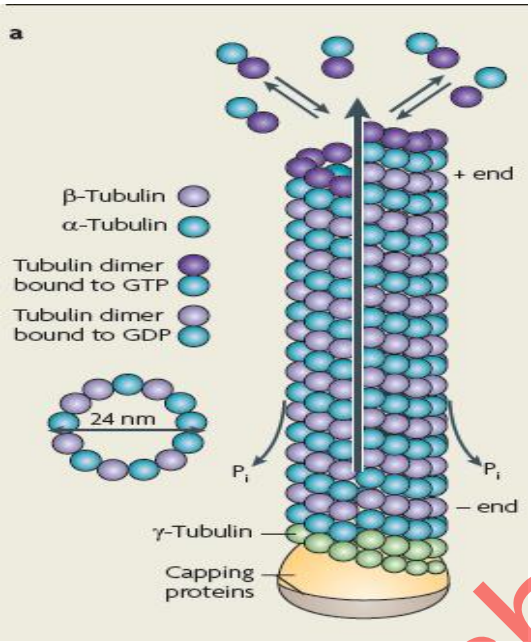
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Microtubules

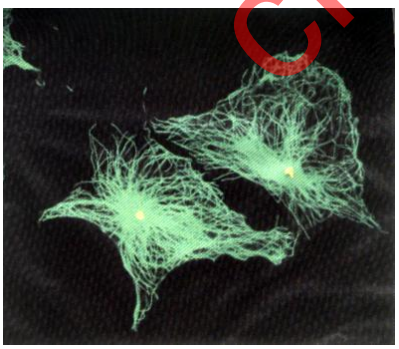
Rigid internal skeleton resist compression. Tracks along for motor proteins. Tubulin is a dimer of α-tubulin and β-tubulin. The two ends different plus (+) end & minus (–) end (cat Tail & dynamic).

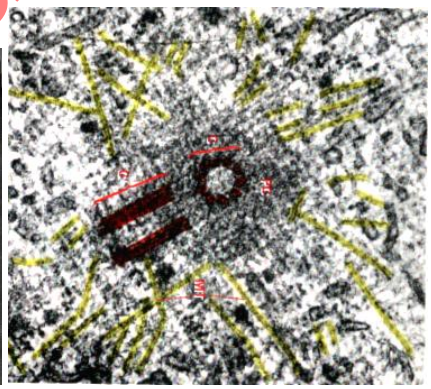
Microtubules 13



Centrosome

Microtubule organizing center





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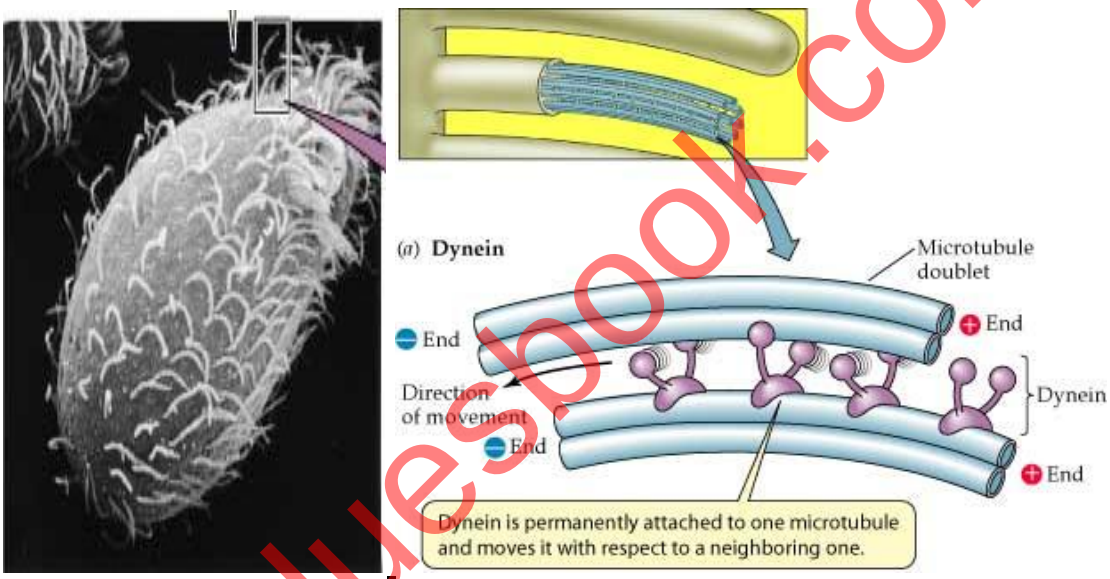
Topic 23 Motor Protiens

Dynein & Kinesin

Cargo & Motor Domain



Cilia and Flagella



ECM

Holds cells in tissues. Physical properties of tissues cartilage, skin. Orients cell movements during development/injury. Chemical signaling from one cell to another.



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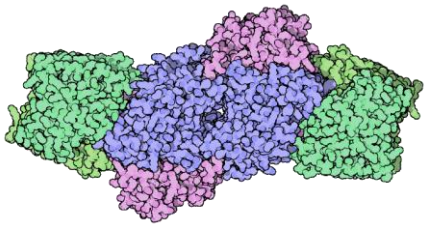
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Topic 24 The Miracle Workers

$N_2 + 3H_2 \rightarrow 2NH_3$

The reaction is carried out under conditions of 150-250 atmospheres (atm), 450-500 °C; resulting in a yield of 10-20%

$N_2 + 8H^+ + 8e^- + 16ATP \rightarrow 2NH_3 + H_2 + 16ADP + 16\ Pi$



Nitrogenase

Energy Conversions & Enzymes

The Miracle Workers

One of biggest miracles of life is that it can perform slow reactions and the ones requiring extreme conditions, quickly and at physiological conditions.

Topic 25 Energy

Capacity to do work is energy.

Potential energy is the energy of state or position; it includes the energy stored in chemical bonds.

Kinetic energy is the energy of motion (and related forms such as electric energy, light, and heat).

Energy Conversions

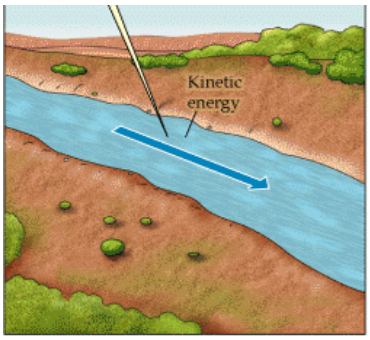
Efficiency Not 100%

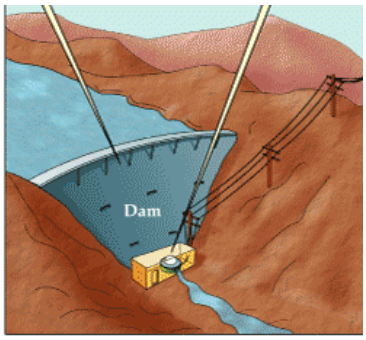
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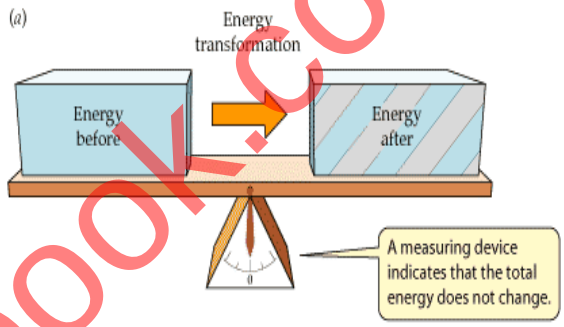


The Laws of Thermodynamics

The First Law

Energy cannot be created or destroyed.

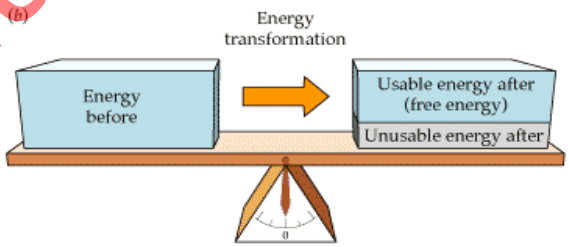
The First Law of Thermodynamics. The total amount of energy before a transformation equals the total amount after a transformation. No new energy is created, and no energy is lost.

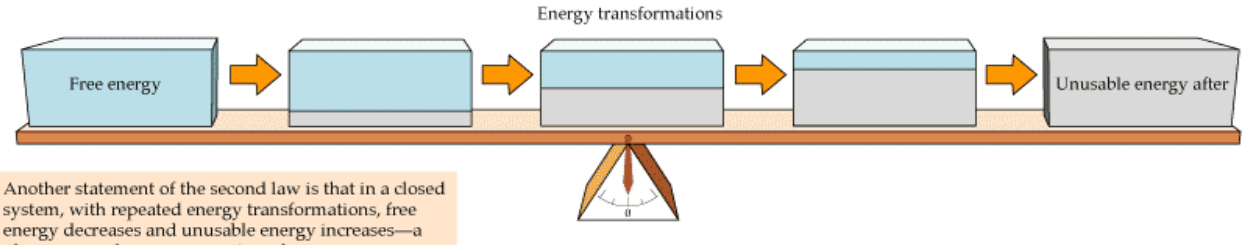


The Second Law

The quantity of energy available to do work (free energy) decreases and unusable energy (associated with entropy) increases.

The Second Law of Thermodynamics. Although a transformation does not change the total amount of energy within a closed system, after any transformation the amount of free energy available to do work is always less than the original amount of energy.





Another statement of the second law is that in a closed system, with repeated energy transformations, free energy decreases and unusable energy increases—a phenomenon known as creation of **entropy**.

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Free Energy (ΔG)

The Second Law

Total energy = usable energy + unusable energy

H = G + TS

(H = Enthalpy, G = free energy, T= absolute temperature, S = entropy)

G = H – TS

Topic 26 Free Energy

Free Energy (ΔG)

Absolute G, H or S cannot be measured

Changes in free energy, total energy, and entropy are related by the equation.

ΔG = ΔH – TΔS

ΔG (reaction) = G (products) – G (reactants)

ΔH (reaction) = H (products) – H (reactants)

ΔS (reaction) = S (products) – S (reactants)

ΔG of Combustion

Reactants	**Products**		
$CH_4 + 2O_2$	**$CO_2 + 2H_2O$**		
H = G + TS	**H = G + TS**	**500 = 400 + 100**	**200 = 100 + 100**

Reactants → Products

G = H – TS → G = H – TS

400 = 500 - 100 → 100 = 200 - 100

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$\Delta G_{(reaction)} = G_{(products)} - G_{(reactants)}$

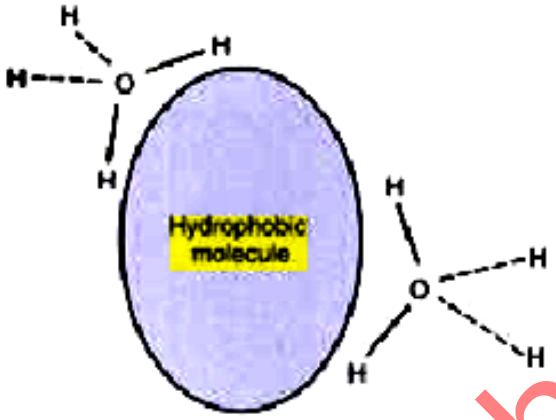
- 300 = 100 - 400

ΔG/ ΔH = - 300

Hydrophobicity Counterintuitive

????

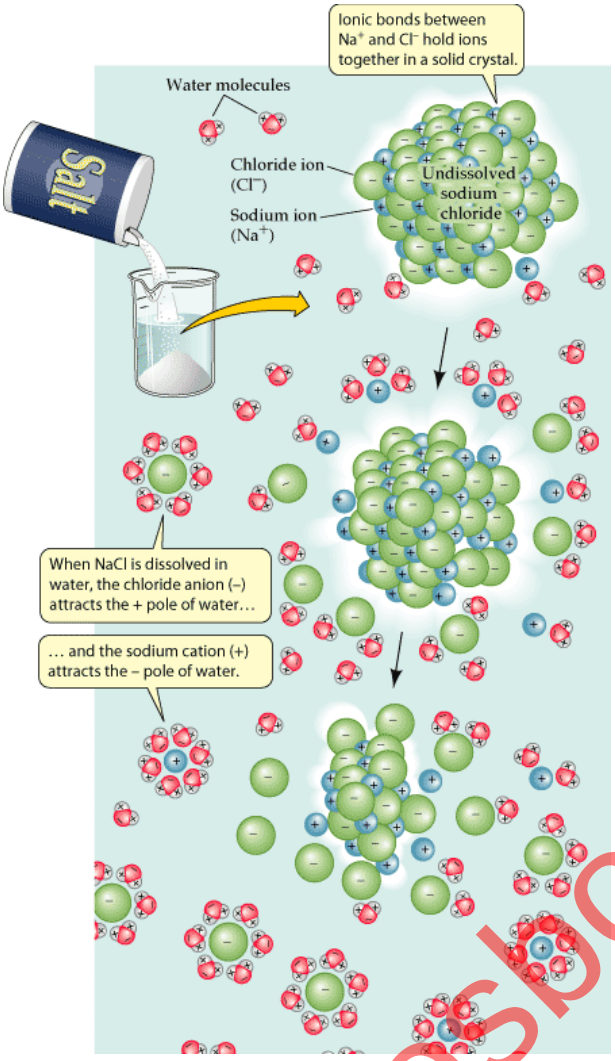
H_2O adjacent to hydrophobic molecule are restricted to form hydrogen bonds with other water molecules.



Bonding (hydrogen) also reduces energy state.

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Free Energy (ΔG)

+ve or -ve

If ΔG is -ve energy is released

If ΔG is +ve free energy is required

In living systems magnitude and sign of ΔG can depend significantly on changes in entropy ΔS

ΔS = +ve in hydrolysis

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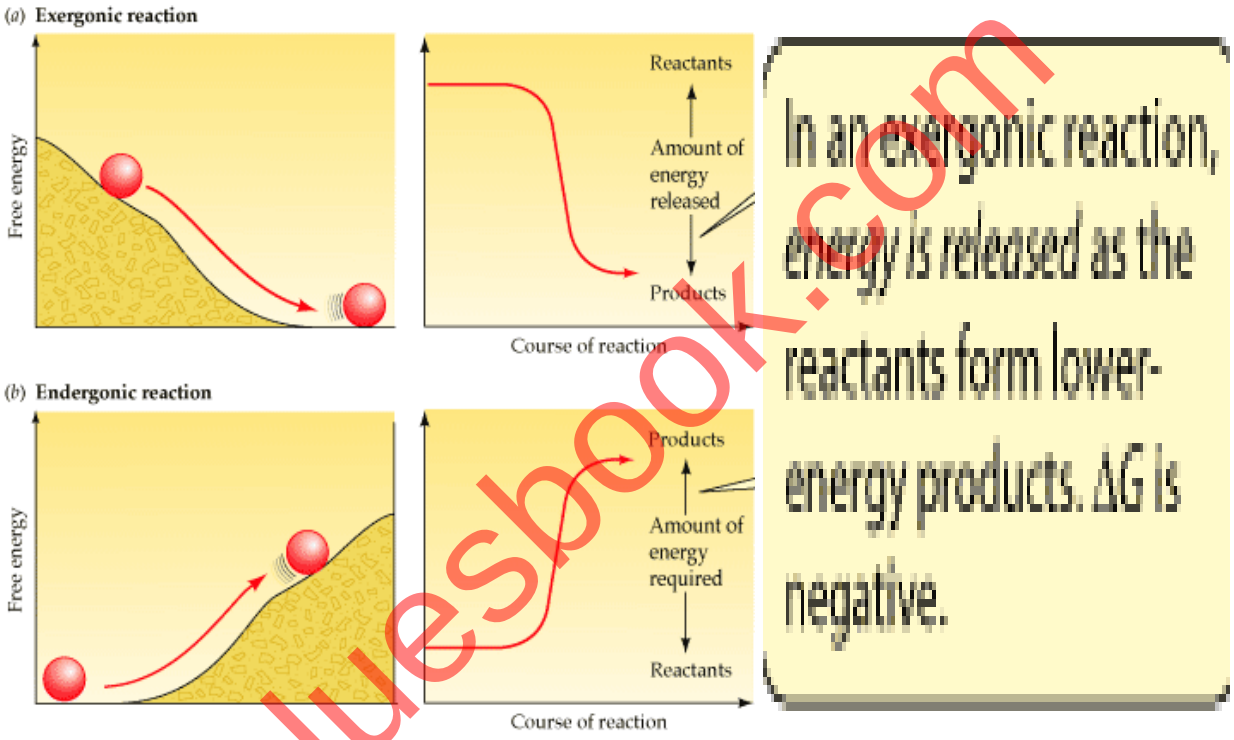
Topic 27 Exergonic & Endergonic Reactions

Exergonic & Endergonic Reactions

ΔG $^{+}$ve or $^{-}$ve

Endergonic reactions take up free energy and have a positive ΔG. Endergonic reactions proceed only if free energy is provided

Exergonic reactions release free energy and have a negative ΔG.



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Reversible Reactions

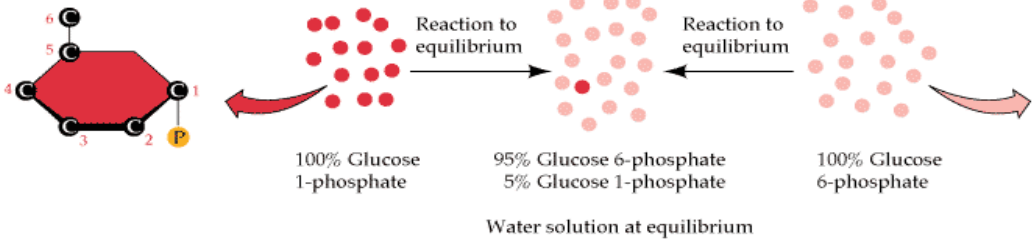
Equilibrium: Point at which the forward and reverse reactions proceed at the same rate. The change in free energy (ΔG) of a reaction determines its point of chemical equilibrium. For exergonic reactions, the equilibrium point lies toward completion (the conversion of all reactants into products).

A→ ←B

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At equilibrium

An amount reactant and product molecule stays constant. Although individual moles molecule keep changing their states

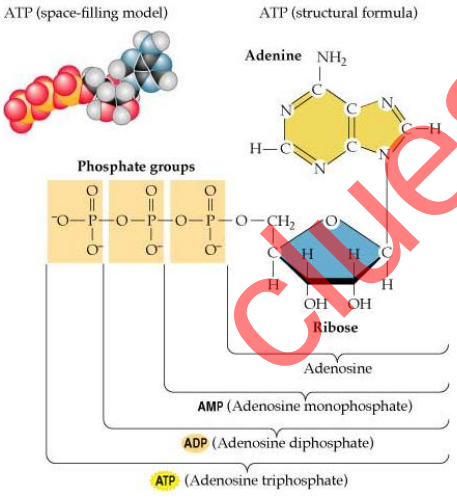
$\Delta G = 0$

Topic 28 ATP Hydrolysis

ATP Hydrolysis

Energy Currency

Hydrolysis of ATP releases a relatively large amount of free energy (-12 kcal/mol).



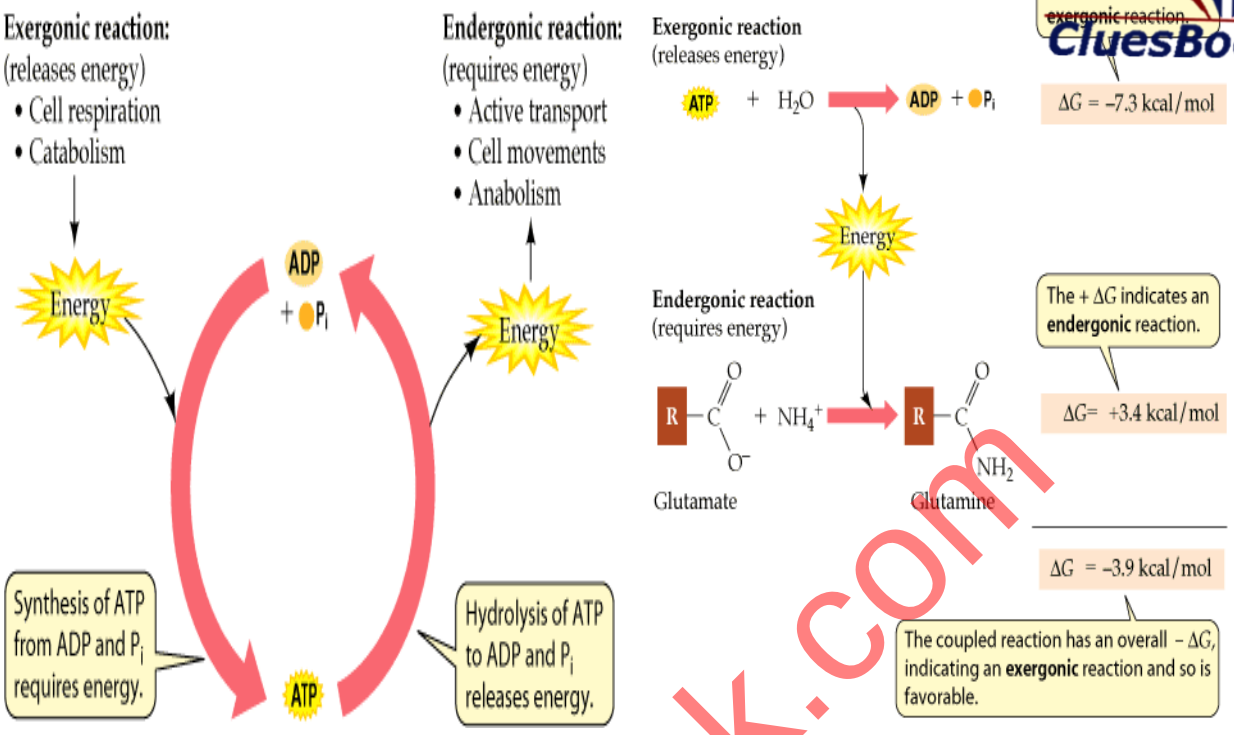


Coupling

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Speed of a Reaction

ΔG Independent

50% 600 poly arginine degradation in 7years Carboxypeptidase takes half a second. Reactants need to cross an energy barrier. Small amount of added energy (Activation Energy E_a) e.g. CH4 + O2

Topic 29 Enzymes The Biological Catalyst

Enzymes

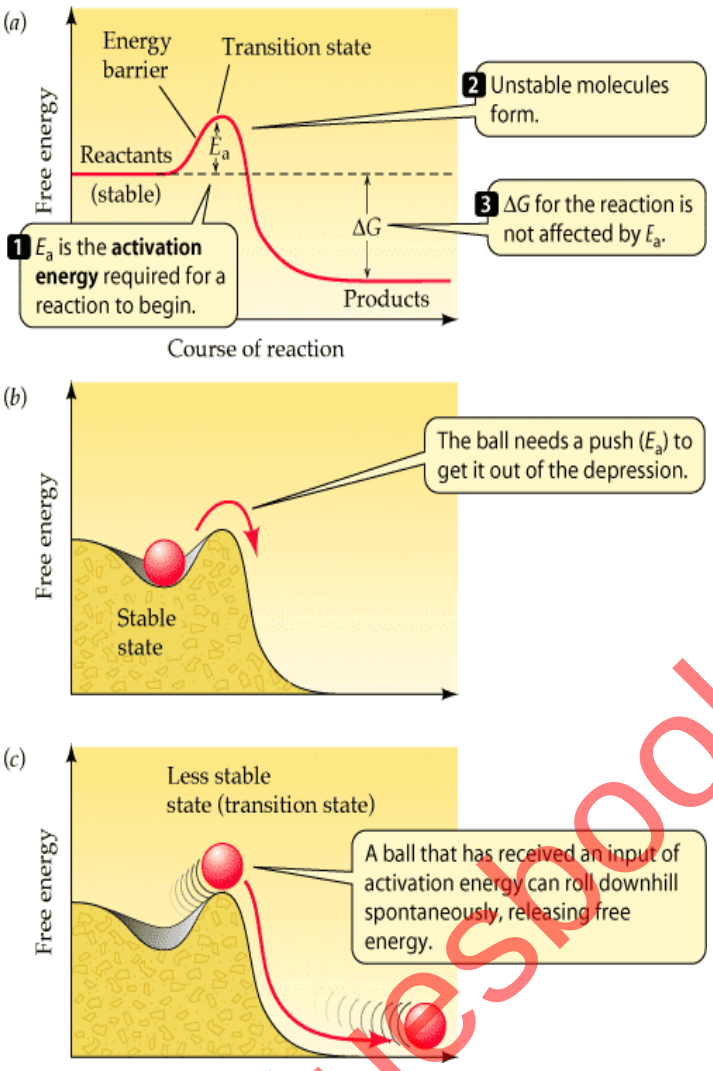
The more ¯ve ΔG is, the further the reaction proceeds to completion. ΔG does not tell us the rate of a reaction (the speed at which it moves towards equilibrium). Most Enzymes are proteins (ribozymes are RNA)

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Activation Energy



At normal temperatures few molecules have enough kinetic energy for this transformation.

Topic 30 Enzymes Mechanisms

Enzymes Specificity

Reactants (substrates) bind the particular (active) site of the respective enzyme.

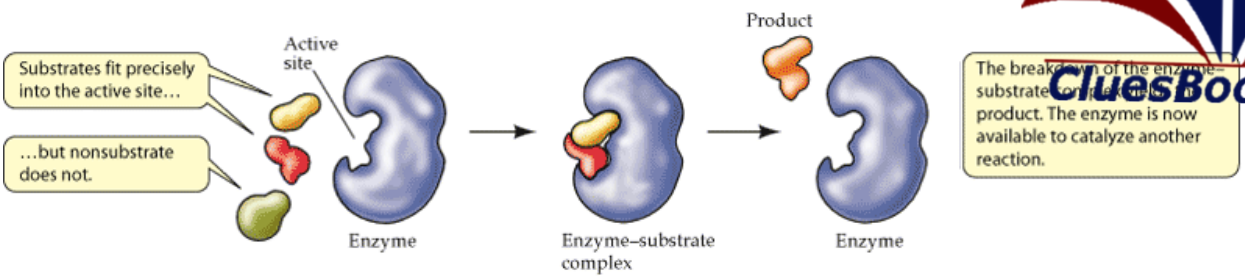
The active site = catalytic + binding sites

The specificity of this binding is a function of three-dimensional shape, structure and environment of its active site.

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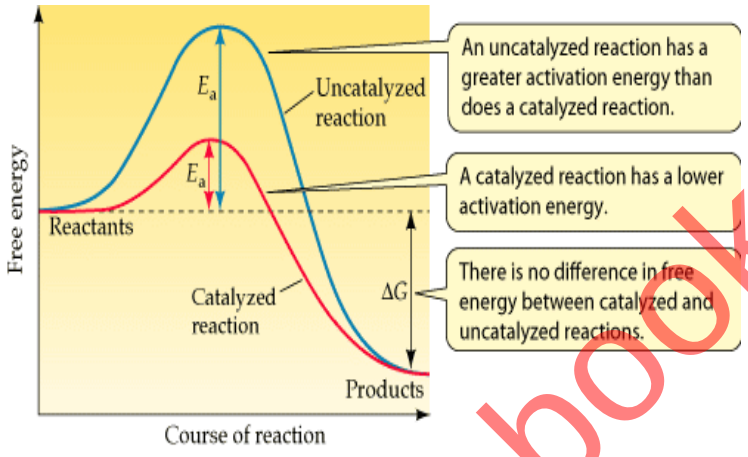
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Enzymes Lower the Energy Barrier

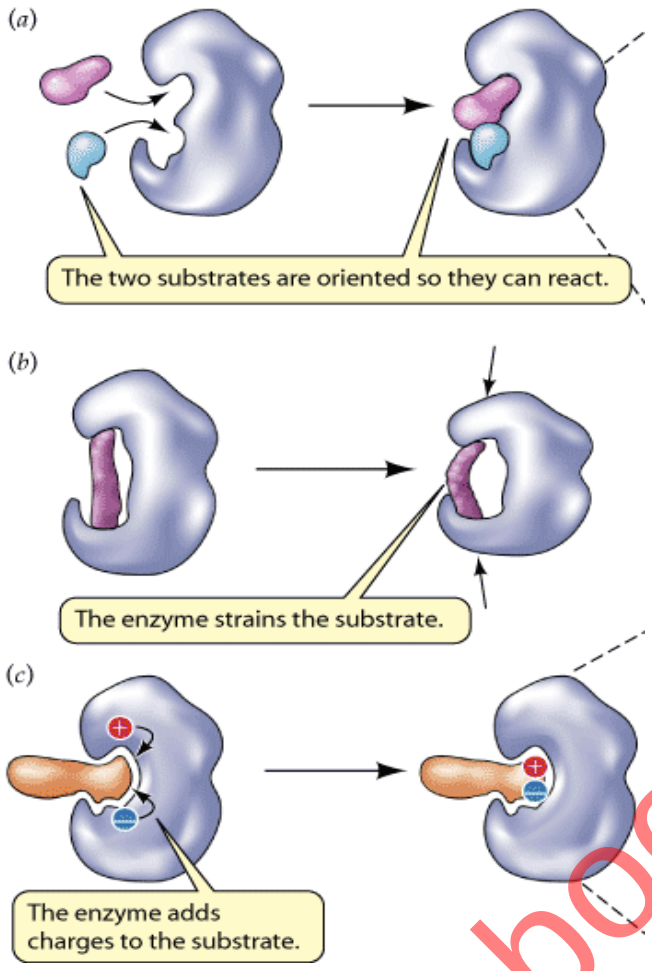
E_a Enzymes Lower the Energy Barrier (forward and reverse reaction). Equilibrium and ΔG are not changed.



How do Enzymes Work?

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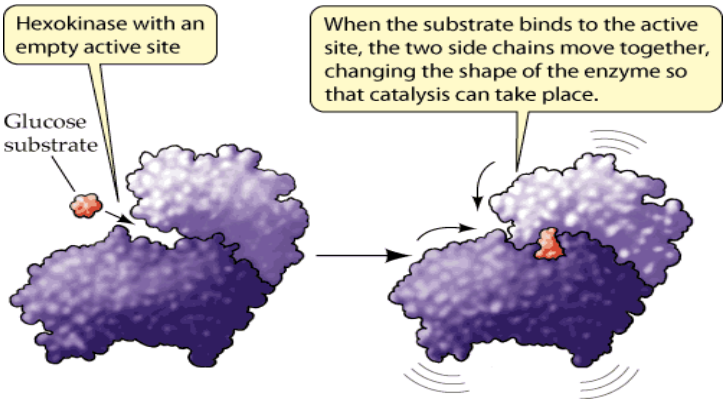
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Induced Fit

Conformational Change Upon binding to substrate, some enzymes change shape, facilitating catalysis. The shape changes (Induced fitting) modifies active site and exposes/ aligns those regions that perform catalysis.

Water exclusion at active site results in transfer of Pi from ATP to glucose preventing APT



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Topic 31 Enzyme Helpers

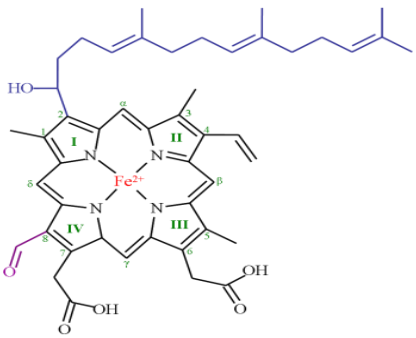
6.1 A Few Examples of Nonprotein "Partners" of Enzymes

TYPE OF MOLECULE	ROLE IN CATALYZED REACTIONS
Cofactors	
Iron (Fe^{2+} or Fe^{3+})	Oxidation/reduction
Copper (Cu^{+} or Cu^{2+})	Oxidation/reduction
Zinc (Zn^{2+})	Helps bind NAD
Coenzymes	
Biotin	Carries $—COO^{-}$
Coenzyme A	Carries $—CH_2—CH_3$
NAD	Carries electrons
FAD	Carries electrons
ATP	Provides/extracts energy
Prosthetic groups	
Heme	Binds ions, O_2, and electrons; contains iron cofactor
Flavin	Binds electrons
Retinal	Converts light energy

Cofactors: Inorganic ions (Cu, Zn & Fe) bound to some enzymes and essential to their function.

Coenzymes: Carbon containing molecules bind enzymes temporarily chemically changed during reaction (substrate).

Prosthetic groups: non-protein component permanently bound to enzymes (heme group), without its prosthetic group apoprotein vs. holoprotein.

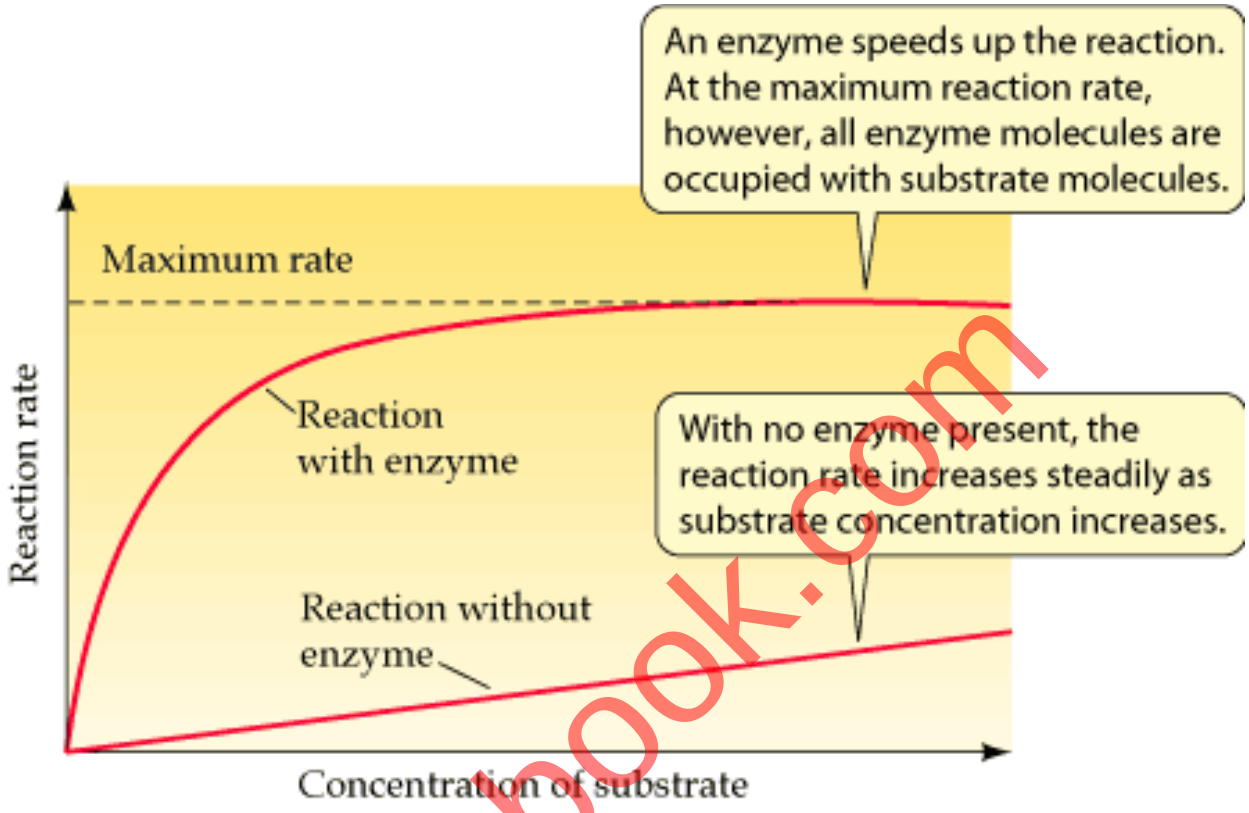


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Topic 32 Enzyme Regulation

Substrate Concentration



Enzyme Regulation

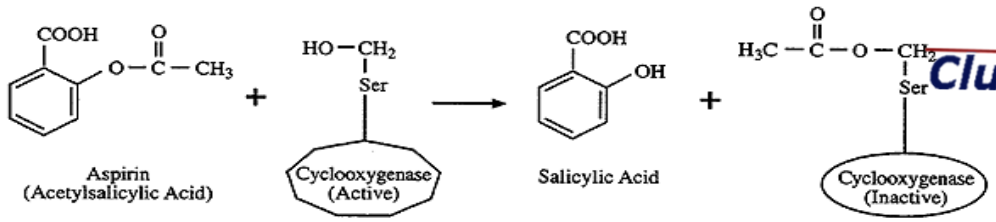
The Inhibitors: Some factors can activate enzymes and other can deactivate them (inhibit their function).

1. Irreversible inhibitors
2. Competitive inhibiters
3. Noncompetitive inhibiters

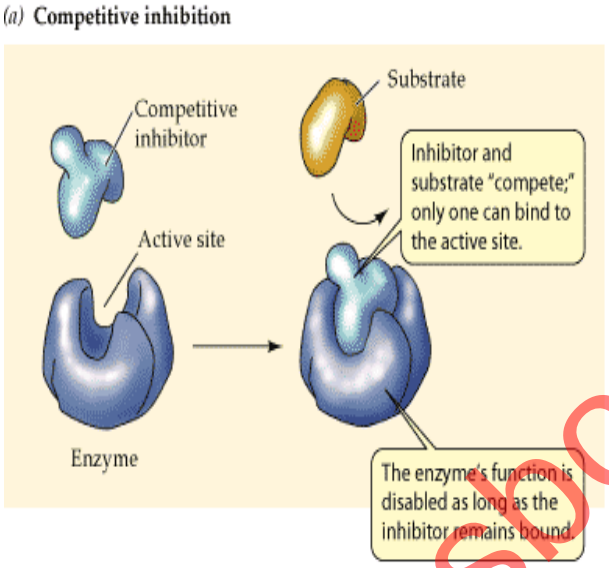
1. **Irreversible inhibitors :** Permanently inhibit enzymes
   Function generally by modifying the active site.

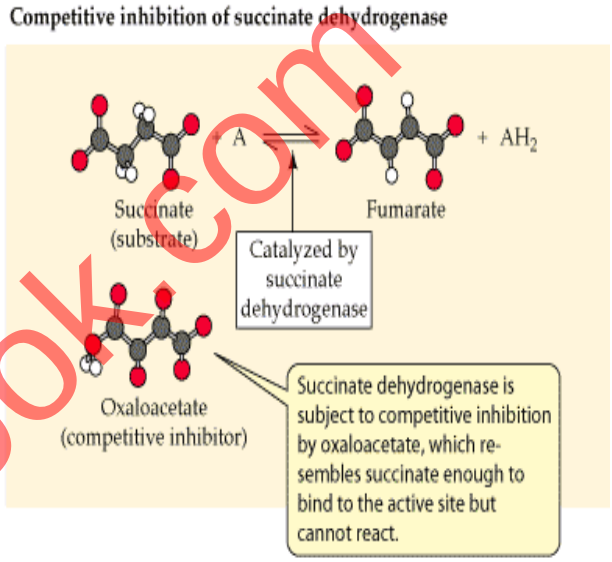
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2. **Reversible inhibitors**
3. **Competitive inhibiters** are similar to the natural substrate (bind the active site) however they are different enough that no reaction is catalyzed.





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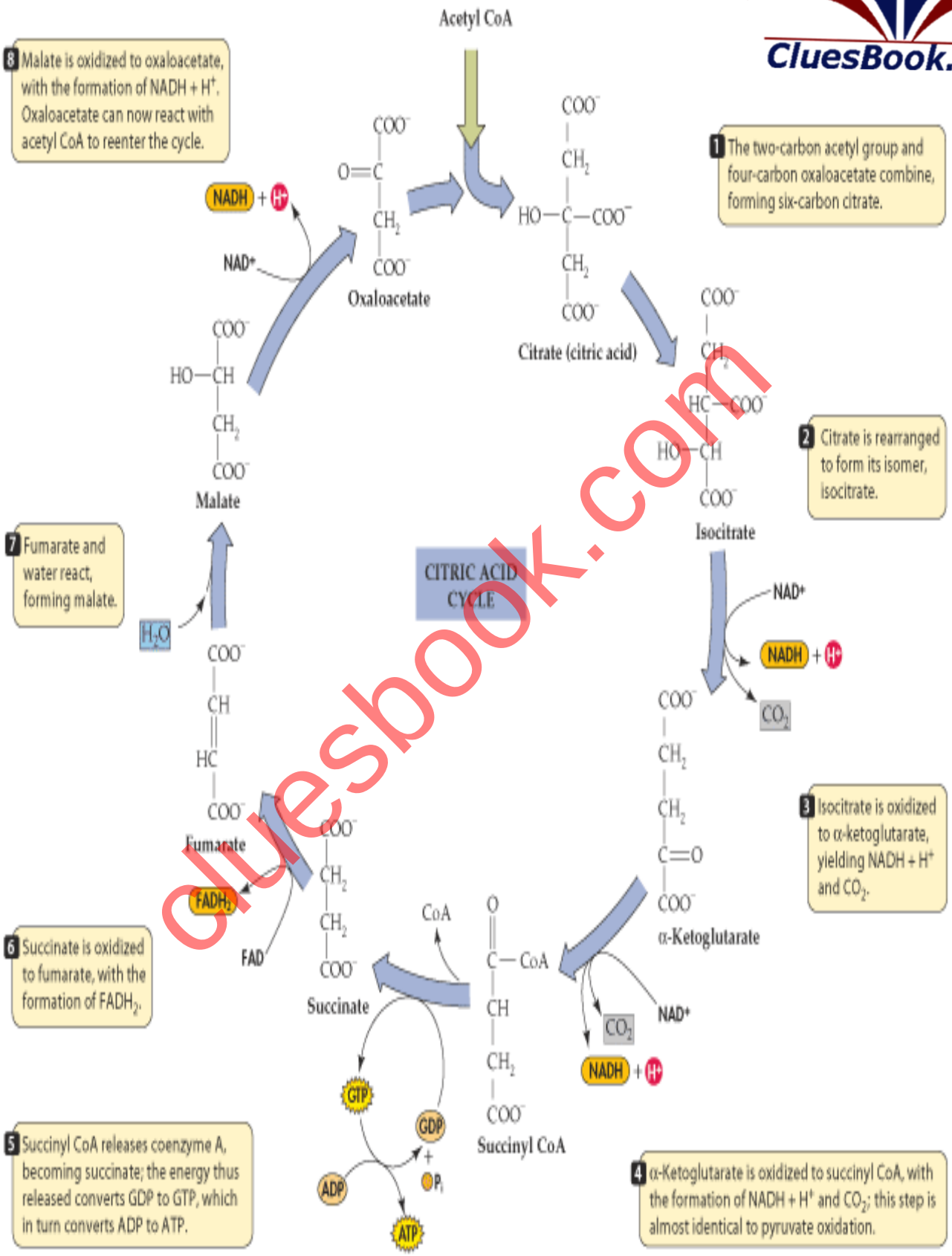
The Citric Acid Cycle

Feed Back/End Product Inhibition

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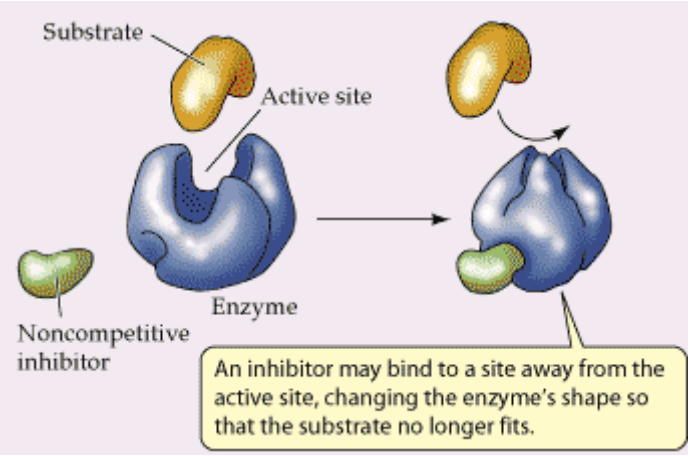


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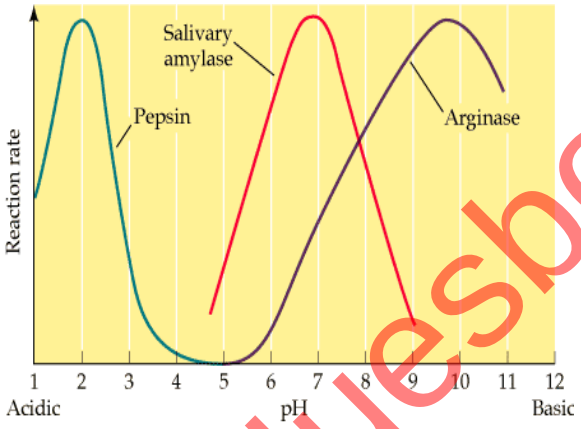
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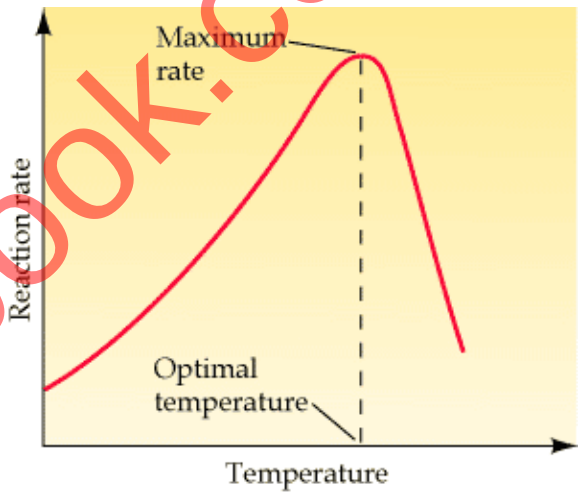
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Noncompetitive inhibiters : Bind the enzyme at nonactive site cause a conformational change prevents enzyme from binding its substrate.



Physical Factors
Influence Enzymes





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